When Accuracy Saturates: A Feature-Node Graph Representation for Deployable and Transportable Cardiovascular Risk Prediction

First A. Author, *Student Member, IEEE*
Department of Computer Science and Engineering, Institution Name, City 000000, Country
E-mail: author@institution.edu

***Abstract*—On small clinical tabular cohorts, competing models frequently reach statistically indistinguishable accuracy, so accuracy alone is a weak basis for choosing between them. We evaluate a *feature-node* graph formulation—each patient is an independent graph whose thirteen nodes are clinical features, with edges from training-fold Pearson dependence thresholded at τ = 0.15 and augmented with a minimum spanning tree—against three properties that should govern adoption when accuracy is tied: representation, deployability and transportability. Under a leakage-controlled five-fold protocol shared with six classical baselines (Logistic Regression, Random Forest, Gradient Boosting, a Multilayer Perceptron, XGBoost and LightGBM) over three seeds, a two-layer graph convolutional network is statistically indistinguishable from every one of them, including the two modern gradient-boosted-tree models (McNemar p = 0.70 against Logistic Regression, p = 0.77 against Random Forest, p = 0.68 against Gradient Boosting, p = 0.68 against the Multilayer Perceptron, p = 1.00 against XGBoost, p = 0.66 against LightGBM) on 303 patients; we report this parity as a finding, not an improvement. The formulation differs operationally: it is inductive, scoring single patients with no cohort present, reproducing batch predictions to 3×10^{-8} in 2.6 ms from an 11.4 KB, 1,281-parameter model, whereas the conventional patient-similarity GNN cannot score an isolated patient at all and collapses to a single-class predictor on one external cohort. Because three of the thirteen features (`ca`, `thal`, `slope`) are not recorded in the other cohorts, cross-cohort transport uses a prespecified eight-feature version of the same architecture, retrained once on Cleveland and evaluated once per cohort; on that eight-feature model the graph attains higher ROC-AUC in all six model–cohort comparisons and retains calibration under shift where the linear baseline collapses (0.091 versus 0.318 expected calibration error on Hungarian); bootstrap intervals establish the advantage against Logistic Regression on two of three cohorts and against Random Forest on none. Ablation localises what matters to graph *construction*—selective sparsity outperforms random and fully connected topologies—rather than to the convolution operator, which is statistically irrelevant. Code, seeds and all result tables are released.**

***Index Terms*—Cardiovascular disease, clinical tabular data, cross-cohort transport, distribution shift, explainable artificial intelligence, feature graph, graph neural networks, inductive inference, model transportability, representation learning.**

I. Introduction

This work received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Corresponding author: First A. Author (e-mail: author@institution.edu).

Cardiovascular disease (CVD) remains the leading cause of mortality worldwide [1], and the early identification of at-risk patients from routinely collected clinical measurements is of considerable importance. Machine-learning models have been applied extensively to this task, yet two obstacles limit their clinical adoption: many high-performing models treat features independently and therefore ignore the well-documented interactions among cardiovascular risk factors, and the most accurate models are frequently opaque, which is problematic in a domain where accountability is essential [2], [3].

Graph neural networks [4], [5] offer a natural mechanism for modelling relational structure. In clinical applications they are most commonly instantiated as patient-similarity graphs, where nodes are patients and edges connect clinically similar individuals. Such formulations are transductive, require a cohort-level graph at inference time, and do not expose the interactions *between features* that clinicians reason about. In this work we study the complementary *feature-node* formulation: every patient is an independent graph whose nodes are the thirteen clinical features and whose fixed topology encodes feature–feature dependence. Message passing then propagates information along data-derived feature associations, and the resulting representation is inductive and inspectable.

Position of this paper. We deliberately do not frame this work as a pursuit of higher accuracy. On a cohort of 303 near-balanced patients every competently tuned model saturates at ROC-AUC ≈ 0.90, and differences at that ceiling are dominated by splitting noise; a further decimal place of accuracy would be neither reproducible nor clinically meaningful. Nor do we claim architectural novelty: our own ablation shows that exchanging the convolution operator for GAT, GraphSAGE or GIN changes no metric significantly (Section VII-F). Framing the contribution in either of those terms would misdescribe what the evidence supports.

Instead we evaluate the feature-node formulation against the three properties that we argue should govern the adoption of a clinical prediction model when accuracy is tied:

- **P1 Representation.** Does the model expose the structure of the problem in a form a clinician can inspect and interrogate—an explicit map of how risk factors relate, and explanations that independent attribution methods agree upon?
- **P2 Deployability.** Can the model score a single patient, in

isolation, cheaply and without infrastructure—that is, is it inductive, compact and free of any requirement for a cohort at inference time?

P3 **Transportability.** When moved to a population it was not developed on, does it degrade gracefully, and does it fail safely rather than collapsing to a degenerate predictor?

Accuracy is reported throughout, but as a *control*: our claim is that the feature-node representation is competitive, not superior, and we subject that parity to significance testing rather than obscuring it. The novelty we claim is deliberately narrow: not a new convolution operator (Section VII-F shows GCN, GAT, GraphSAGE and GIN do not significantly differ), but the feature-node formulation itself together with the evaluation framework that stress-tests it on the three properties below, each measured rather than asserted. Our contributions follow these three properties directly.

C1 **Representation.** We formulate clinical tabular data as a feature-node graph in which correlation-derived edges are augmented by a minimum spanning tree, constructed inside each fold without information leakage and validated spectrally before any training occurs (Section IV). We characterise this construction rather than merely asserting it: we show that the MST is a *connectivity safeguard* that contributes no edges at the operating threshold and only becomes active for sparser graphs, that in the regime where it is active there is suggestive evidence that an MST-supported sparse topology is more favourable under distribution shift, even though it leaves in-distribution accuracy unchanged (Section VII-M), and that a global correlation cut-off and an adaptive per-node k-NN rule of comparable density are statistically indistinguishable after correction for multiple comparisons (Sections VII-G, VII-H). We then establish that the representation yields convergent and clinically coherent explanations through a multi-method explainability evaluation across eight metrics, including deletion/insertion curves and cross-method agreement (Section VII-J).

C2 **Deployability.** We compare the formulation directly against the conventional patient-similarity GNN and quantify the operational cost of transductivity: that baseline cannot score an isolated patient and must rebuild a graph over any cohort it is given, whereas the feature-node model is inductive, compact and requires no cohort at inference time (Sections VII-C, VII-D).

C3 **Transportability.** We validate on three other sites within the same UCI Heart Disease collection under strict Cleveland-only fitting, with paired bootstrap intervals that propagate both sampling and initialisation uncertainty, showing a partial but consistent transport advantage and a difference in failure mode (Section VII-L).

Underpinning all three is a leakage-controlled protocol in which the GCN and six classical baselines—including two modern gradient-boosted-tree models—share identical stratified cross-validation folds, in-fold scaling and inner-validation threshold tuning (Section VI); a comprehensive ablation that localises what matters to graph *construction* rather than architecture (Section VII-F); uniform Holm–Bonferroni control of multiplicity wherever a family of hypotheses is tested; and full public release of code, fixed seeds and result tables. Together these establish accuracy parity as a rigorously supported null result rather than an unexamined assumption. An overview of the methodology is shown in Fig. 1.

II. Related Work

Graph neural networks. The graph convolutional network of Kipf and Welling [4] popularised spectral-inspired neighbourhood aggregation; subsequent architectures introduced attention [6], inductive sampling [7], and provably expressive aggregation [8]. We adopt the GCN for its simplicity and because the small, fixed feature graph does not require the additional capacity of attention-based variants; the choice is validated empirically in our ablation.

GNNs for clinical data. Most clinical GNNs are patient-similarity graphs, in which nodes are patients and edges encode cohort-level similarity [9]; recent surveys of graph representation learning in biomedicine and healthcare confirm that this patient-graph formulation remains the dominant paradigm across applications from disease progression to multi-omics integration [10], [11]. The feature-node view used here—in which every patient is an independent graph over *features* rather than a node in a graph over *patients*—is comparatively under-explored in the clinical literature specifically, even though it has begun to appear in the general tabular-learning literature outside medicine: T2G-FORMER learns a relation graph over tabular features to promote heterogeneous feature interaction [12], and a contextual graph-embedding approach similarly organises tabular columns as graph nodes for deep learning on non-relational data [13]. A recent taxonomy of graph neural networks for tabular data explicitly separates "instance graphs" (patient-similarity, the dominant clinical case) from "feature graphs" (our formulation) as two distinct and under-balanced branches of the field, with the survey's own taxonomy counting substantially more instance-graph than feature-graph clinical applications [14]. Competing approaches to tabular relational structure include attention-based tabular transformers, which learn feature interactions implicitly through self-attention rather than an explicit, inspectable graph, and which recent surveys note trade interpretability for flexibility relative to graph-structured alternatives [15]; and tabular foundation models such as TabPFN, which achieve strong accuracy on small tabular cohorts through in-context learning but, like attention-based transformers, do not expose an explicit feature-adjacency structure a clinician can inspect [16]. Rather than merely assert the feature-node formulation's advantage over the dominant patient-similarity paradigm, we implement a patient-similarity GNN as an explicit baseline and compare the two formulations both in-distribution and under transport (Sections VII-C and VII-L).

Explainable AI. Post-hoc attribution has become central to trustworthy machine learning [17], and the case for interpretability as a first-class requirement—rather than a post-hoc add-on—in deployed clinical models has been argued at length in recent reviews [18]. Model-agnostic attribution methods such as SHAP remain the most widely adopted explanation

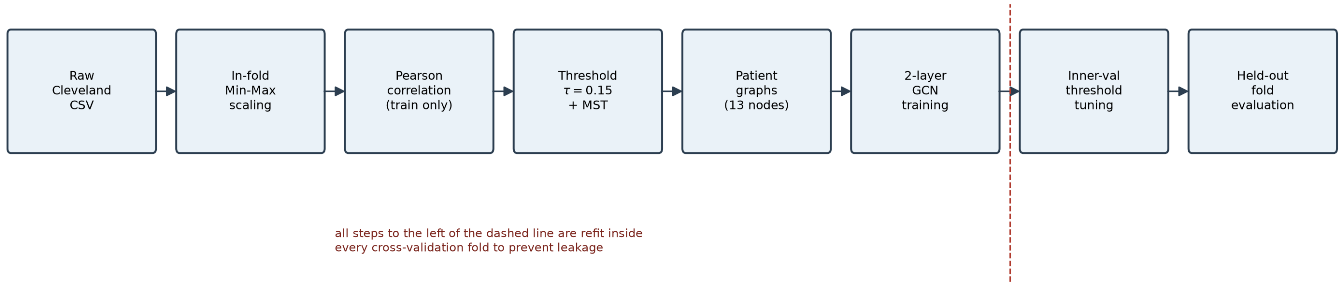

Fig. 1. Overall methodology pipeline. Preprocessing, correlation-graph construction with MST augmentation, patient-graph encoding, GCN training with inner-validation threshold tuning, and post-hoc explanation are all performed inside each cross-validation fold to prevent information leakage.

TABLE I
UCI Cleveland dataset characteristics.

Property	Value
Samples	303
Features	13
Continuous features	5
Categorical features	8
Positive class (disease)	139 (45.9%)
Negative class (no disease)	164 (54.1%)
Missing values	0
Duplicate rows	0

tool in applied biomedical machine learning [19], but they are computed post-hoc on models that do not themselves expose relational structure. We instead employ GNNExplainer [20], which learns compact feature/edge masks directly on the feature graph, together with the gradient-based Integrated Gradients [21] and Saliency [22], and evaluate them with fidelity, sparsity, stability, sensitivity [23] and deletion/insertion analyses.

III. Materials

A. Dataset

We use the UCI Cleveland Heart Disease dataset [24], [25], comprising 303 patients described by thirteen clinical features and a binary target (Table I). The prediction task is binary: presence (1) versus absence (0) of cardiovascular disease. The cohort is nearly balanced (164 versus 139), and there are no missing values in the version used. Feature definitions and clinical semantics are given in Table II. The class distribution motivates our use of threshold tuning and MCC rather than accuracy alone.

B. Preprocessing

The five continuous features are scaled with min–max normalisation; categorical and ordinal features are retained as integer codes. Crucially, the scaler is fit on the training partition of each fold only and then applied to validation and test data, so no test-set statistics influence training.

TABLE II
Clinical feature definitions.

Feature	Description	Type
age	Age in years	continuous
sex	Sex (1 = male, 0 = female)	categorical
cp	Chest pain type (0-3)	categorical
trestbps	Resting blood pressure (mmHg)	continuous
chol	Serum cholesterol (mg/dl)	continuous
fbs	Fasting blood sugar >120 mg/dl (1/0)	categorical
restecg	Resting ECG result (0-2)	categorical
thalach	Maximum heart rate achieved	continuous
exang	Exercise-induced angina (1/0)	categorical
oldpeak	ST depression induced by exercise	continuous
slope	Slope of peak exercise ST segment	categorical
ca	Number of major vessels coloured by fluoroscopy (0-3)	categorical
thal	Thallium stress test result	categorical

IV. Feature-Node Graph Construction

A. Correlation graph

Let $X_{\text{tr}} \in \mathbb{R}^{n\times 13}$ denote the scaled training features of a fold. We compute the Pearson correlation matrix $R \in \mathbb{R}^{13\times 13}$ and create an undirected edge between features i and j whenever $|R_{ij}| \geq \tau$, with the threshold fixed at $\tau = 0.15$. The threshold was selected by the sensitivity analysis of Section VII-F and held constant thereafter. The full-cohort correlation structure is visualised in Fig. 2.

B. Minimum spanning tree augmentation

A purely thresholded graph may be disconnected, which would prevent message passing from reaching isolated features. We therefore compute a minimum spanning tree over the complete weighted graph with edge distance $d_{ij} = 1-|R_{ij}|$ using Prim's algorithm [26], and add any MST edge that is not already present. This guarantees a single connected component while preferring to bridge feature pairs that are as correlated as possible. The resulting canonical graph is shown in Fig. 3, and its statistics—reported as mean±std across folds—appear in Table V.

The augmentation is a *safeguard*, and we are explicit about when it acts. Table III reports the number of bridges it contributes as a function of τ. For $\tau \leq 0.15$ the thresholded graph is already connected in all five folds and the MST adds

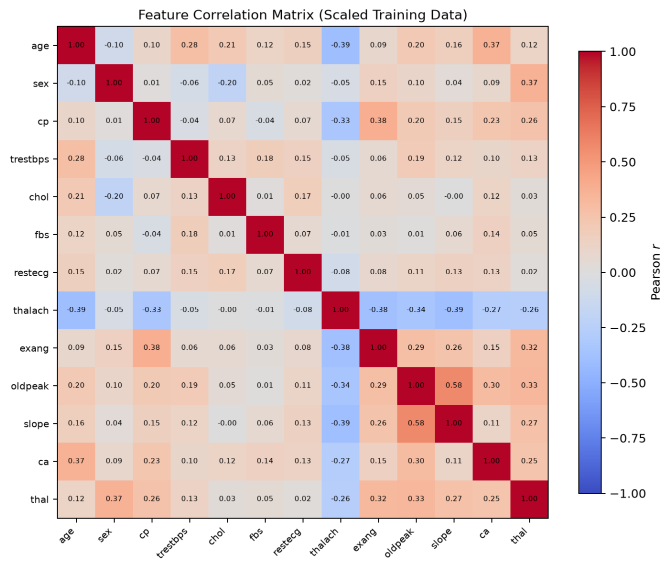

Fig. 2. Pearson correlation matrix of the thirteen clinical features. Edges of the feature graph are instantiated for entries with $|r| \geq \tau = 0.15$; the minimum spanning tree then adds the few bridges required for connectivity.

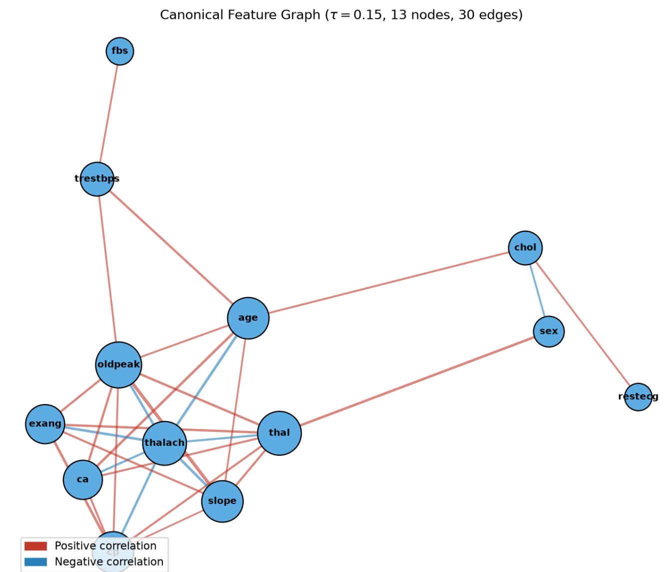

Fig. 3. Canonical feature-node graph ($\tau = 0.15$). Edges are correlation edges (red positive, blue negative) with width proportional to $|r|$. Node colour encodes clinical grouping and node size encodes degree. No MST bridge is drawn because none is required at this threshold: the thresholded graph is already connected in every fold and the augmentation adds no edges (Table III). Bridges appear only for $\tau \geq 0.20$.

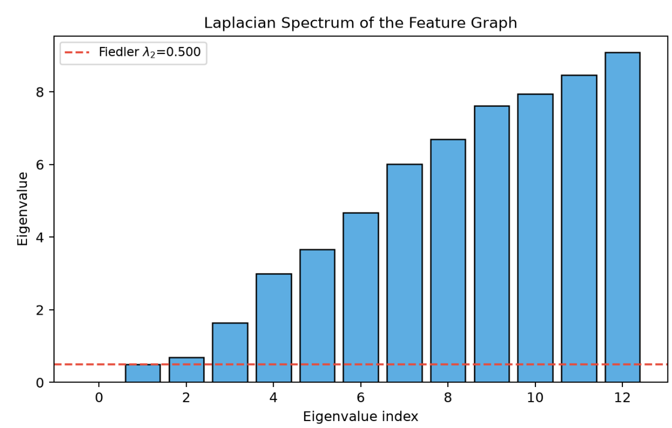

Fig. 4. Laplacian spectrum of the $|r|$-weighted feature graph. The positive Fiedler value λ_2 confirms a single connected component, the structural guarantee provided by the MST augmentation.

no edges whatsoever; from $\tau = 0.20$ onward the thresholded graph is disconnected in every fold and the MST supplies between 1.8 and 9.6 bridges on average. At our operating point of $\tau = 0.15$ the MST is therefore *inert*: it changes nothing. This is worth stating plainly, because it also explains the null result of the "w/o MST" row in the ablation of Section VII-F—at $\tau = 0.15$ the two conditions describe an identical graph, so that row cannot test the MST at all.

To test it we must move to a threshold where it is active. At $\tau = 0.20$ the thresholded graph splits into 2.8 components on average and is connected in 0/5 folds; adding the MST restores a single component in 5/5 folds (Table III). Predictive performance, however, is unchanged: no metric differs significantly between the two conditions (F1 0.7988 vs. 0.7979; ROC-AUC 0.9008 vs. 0.9035; MCC 0.6445 vs. 0.6241; all Holm-adjusted $p = 1.0$, Table IV). The MST is thus a *robustness mechanism that restores connectivity without improving internal accuracy*, and we retain it on that basis rather than as a source of predictive gain. We emphasise *internal* here because this conclusion is drawn entirely within the development cohort. Whether connectivity matters once the model is moved to a different cohort is a separate question that no in-distribution experiment can answer; we return to it in Section VII-M, where the topologies are compared under distribution shift.

C. Spectral and centrality validation

Before training we validate the graph spectrally. The Laplacian is formed with absolute edge weights to preserve positive semi-definiteness, and its second-smallest eigenvalue (the Fiedler value [27]) is strictly positive in every fold (Fig. 4), confirming connectivity. Degree centrality (Fig. 5) identifies `oldpeak`, `thalach` and `thal` as hub features; these coincide with the strongest target correlations, providing an independent, model-free indication that the graph concentrates clinically salient variables.

D. Patient-graph encoding

Every patient is encoded as a graph with thirteen nodes sharing the fold's edge index; the node feature vector is the patient's scalar value for each clinical feature, i.e. $x \in \mathbb{R}^{13\times 1}$. Only the node values vary across patients; the topology is common within a fold.

V. Model

The classifier (Fig. 6) stacks two graph convolutional layers [4], each followed by batch normalisation and ReLU, with dropout between layers. A global mean-pooling readout

TABLE III
MST activation on the 13-feature graph. Panel (a) reports where the thresholded graph fragments; panel (b) reports the resulting connectivity.

(a) Fragmentation by τ

τ	Threshold-graph edges	After MST	MST bridges added	Connected before MST
0.05	59.6	59.6	0.0	5/5
0.1	47.2	47.2	0.0	5/5
0.15	34.4	34.4	0.0	5/5
0.2	23.6	25.4	1.8	0/5
0.25	17.2	20.8	3.6	0/5
0.3	13.4	17.2	3.8	0/5
0.4	2.4	12.0	9.6	0/5

(b) Resulting connectivity

Configuration	Edges	Components	Connected folds
Corr only, τ=0.15	34.4 ± 2.5	1.0	5/5
Corr+MST, τ=0.15	34.4 ± 2.5	1.0	5/5
Corr only, τ=0.20	23.6 ± 1.0	2.8	0/5
Corr+MST, τ=0.20	25.4 ± 1.0	1.0	5/5

TABLE IV
Graph connectivity across the τ regime. Panel (a) reports performance by topology and threshold; panel (b) reports the corresponding paired tests.

(a) Performance by τ

Configuration	Accuracy	F1	ROC-AUC	MCC	Recall	Specificity
Corr only, τ=0.15	0.8032 ± 0.0787	0.7936 ± 0.0692	0.9027 ± 0.0437	0.6164 ± 0.1468	0.8105 ± 0.0825	0.7974 ± 0.1464
Corr+MST, τ=0.15	0.8109 ± 0.0652	0.7971 ± 0.0597	0.9051 ± 0.0413	0.6280 ± 0.1240	0.8011 ± 0.0748	0.8196 ± 0.1208
Corr only, τ=0.20	0.8218 ± 0.0583	0.7988 ± 0.0725	0.9008 ± 0.0406	0.6445 ± 0.1160	0.7841 ± 0.1068	0.8537 ± 0.0648
Corr+MST, τ=0.20	0.8076 ± 0.0770	0.7979 ± 0.0705	0.9035 ± 0.0426	0.6241 ± 0.1425	0.8175 ± 0.0856	0.7992 ± 0.1382

(b) Paired tests

Comparison	Metric	Mean A	Mean B	Δ (B−A)	Wilcoxon p	Holm p	Significant (Holm)
Corr only, τ=0.15 vs Corr+MST, τ=0.15	F1	0.7936	0.7971	0.0035	0.4652	1.0	no
Corr only, τ=0.15 vs Corr+MST, τ=0.15	ROC-AUC	0.9027	0.9051	0.0023	0.0431	0.2586	no
Corr only, τ=0.15 vs Corr+MST, τ=0.15	MCC	0.6164	0.628	0.0116	0.4652	1.0	no
Corr only, τ=0.20 vs Corr+MST, τ=0.20	F1	0.7988	0.7979	-0.0009	1.0	1.0	no
Corr only, τ=0.20 vs Corr+MST, τ=0.20	ROC-AUC	0.9008	0.9035	0.0027	0.2094	1.0	no
Corr only, τ=0.20 vs Corr+MST, τ=0.20	MCC	0.6445	0.6241	-0.0205	0.5754	1.0	no

TABLE V
Feature-graph statistics (mean ± std across the 5 CV folds).

Statistic	Value (mean ± std)
Nodes	13
Edges	34.4000 ± 2.4980
Density	0.4410 ± 0.0320
Avg Degree	5.2923 ± 0.3843
Avg Path Length	1.7051 ± 0.0858
Diameter	3.8000 ± 0.4000
Clustering Coeff	0.5494 ± 0.0888
Fiedler Value	0.9529 ± 0.4437

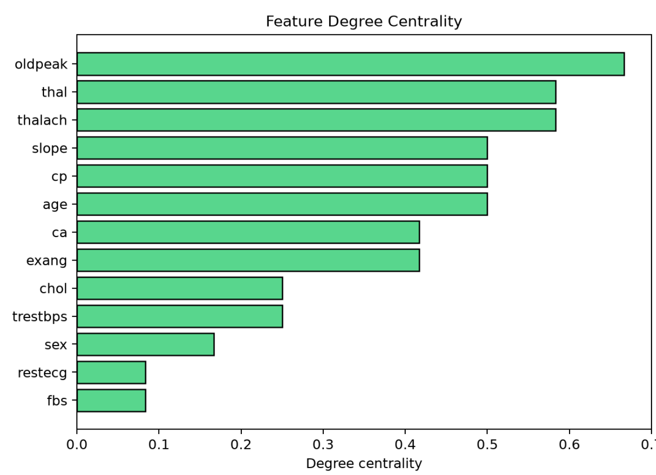

Fig. 5. Degree centrality of the feature nodes. Hub features coincide with the highest target-correlated variables, linking graph structure to clinical relevance.

produces a graph-level embedding that a linear layer maps to a disease probability. Formally, with $H^{(0)} = X$ and normalised adjacency $\hat{A}$,

$$H^{(l+1)} = \mathrm{ReLU}\Big(\mathrm{BN}\Big(\hat{A}H^{(l)}W^{(l)}\Big)\Big), \tag{1}$$

and the prediction is $\hat{y} = \sigma\big(w^\top \mathrm{MeanPool}(H^{(2)}) + b\big)$. Hyper-parameters are listed in Table VI.

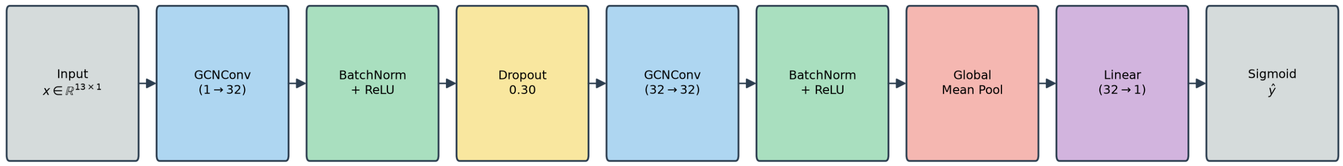

Fig. 6. Two-layer GCN architecture with batch normalisation, dropout, global mean pooling and a sigmoid output head.

TABLE VI
TRAINING AND ARCHITECTURE HYPERPARAMETERS.

Hyperparameter	Value
Node feature dim	1 (scalar clinical value)
# Nodes	13 (clinical features)
Continuous features (scaled)	5
Categorical features (raw)	8
Correlation metric	Pearson
Correlation threshold τ	0.15
Graph augmentation	Minimum Spanning Tree ($d = 1 - \|r\|$)
GCN layers	2
Hidden dimension	32
Normalization	BatchNorm1d
Activation	ReLU
Dropout	0.30
Readout	Global mean pooling
Output head	Linear(32→1) + sigmoid
Loss	BCEWithLogits (balanced)
Optimizer	Adam
Learning rate	1e-3
Weight decay	1e-4
Gradient clipping	max-norm 1.0
Batch size	32
Max epochs	150
Early stopping patience	25 (val loss)
Decision threshold	tuned on inner-validation (F1)
Cross-validation	5-fold Stratified
Inner val split	15% of train fold (stratified)
Scaling	MinMax (continuous cols), fit in-fold
Framework	PyTorch 2.x + PyG 2.7, CPU
Seeds	42, 7, 123

VI. Experimental Setup

A. Protocol

All models are evaluated with five-fold stratified cross-validation, repeated over three random seeds $(42, 7, 123)$. Within each fold a stratified 15% inner-validation split is used for early stopping and decision-threshold selection. Every preprocessing step—scaling and graph construction—is fit on the training partition only; test folds are never used for scaling, graph construction, model selection or threshold tuning. Reported values are mean±standard deviation across folds and seeds.

B. Baselines

Six classical models—Logistic Regression, Random Forest, Gradient Boosting, a Multilayer Perceptron, XGBoost [28] and LightGBM [29]—are trained under *identical* folds with in-fold standardisation and identical inner-validation threshold tuning, ensuring a strictly fair comparison. The two gradient-boosted-tree models are included specifically because they are the strongest tabular baselines in the current literature and were absent from the original baseline suite; their inclusion tests whether a more competitive classical model closes the accuracy-parity gap reported in Section VII.

We additionally implement the *conventional patient-similarity GNN* as a fifth baseline, so that the two graph formulations can be compared empirically rather than only conceptually. Nodes are patients, node features are the patient's scaled clinical values, and edges form a symmetric k-nearest-neighbour graph in feature space; the task is transductive node classification. Depth, hidden width, dropout, optimiser, early stopping, folds and seeds are identical to the feature-node model. To avoid understating the baseline, k is swept over $\{5, 10, 15, 20\}$ and the best value by ROC-AUC is retained ($k = 15$). The scaler is fit on inner-training patients only and the k-NN graph is built from *features alone*—never labels—with the loss masked to training nodes; test-node labels are never observed. Test-node features do participate in message passing, which is inherent to the transductive formulation and is precisely the property we wish to expose.

C. Metrics and statistics

We report accuracy, precision, recall, F1, ROC-AUC, MCC [30] and specificity. Statistical significance is assessed with McNemar's test [31] on the pooled out-of-fold predictions and the Wilcoxon signed-rank test [32] on per-fold F1. Where configurations are compared on the identical (seed × fold) partitions, fold-level metrics are *paired* and the signed-rank test is applied to those pairs.

Wherever a family of hypotheses is tested together—several metrics, several baselines, or several cohorts within one analysis—we control the family-wise error rate with the Holm–Bonferroni procedure [33] and report both the raw and the adjusted p-value, judging significance on the adjusted value. This policy is applied uniformly to every significance table in the paper. It is deliberately *not* applied to the per-seed rows of Table IX, whose rows are replicates of a single hypothesis rather than distinct hypotheses; correcting across replicates would inflate the p-value of one test rather than control a family.

D. Reproducibility

Experiments use Python 3.13, PyTorch 2.x [34], PyTorch Geometric 2.7 [35] and scikit-learn 1.7 [36] on CPU. Random seeds are fixed for NumPy, Python and PyTorch, and per-fold weight initialisation is seeded for determinism. The model has 1,281 parameters; wall-clock training time and peak memory are reported with the released code.

Data and code availability. The UCI Cleveland, Hungarian, Switzerland and Long-Beach-VA heart-disease cohorts are publicly available from the UCI Machine Learning Repository [24], [25]. All experiment code, fixed seeds, generated result tables and figure-producing scripts are released at https://github.com/USERNAME/REPOSITORY. Every table and figure in this paper is regenerated end-to-end by that code; no result is transcribed by hand.

VII. Results and Discussion

A. Predictive performance

Table VII reports the main comparison, which includes gradient-boosted-tree baselines (XGBoost, LightGBM) in addition to Logistic Regression, Random Forest and an MLP, all under the identical 5-fold × 3-seed protocol with in-fold scaling and inner-validation threshold tuning. The feature-node GCN attains the highest mean ROC-AUC (0.906) and specificity (0.824) among all seven models; Logistic Regression is marginally ahead on accuracy (0.818 vs. 0.813), F1 (0.810 vs. 0.799) and MCC (0.646 vs. 0.632), and Random Forest is comparable on ROC-AUC (0.904) but trails on the remaining metrics. The two boosted-tree baselines do not close the gap either: XGBoost reaches ROC-AUC 0.885 ± 0.033 and LightGBM 0.882 ± 0.036, both below every one of the three leading models on this metric, though still within a comparable range on accuracy and F1. All differences among the leading models are within one standard deviation of each other. Pooled out-of-fold point metrics, which aggregate every held-out prediction into a single estimate rather than averaging fold-wise, are reported in Table VII. They agree closely with the fold-averaged values on the ranking metrics (ROC-AUC differs by at most 0.005 for every model), while threshold-dependent metrics differ more for the tree ensembles (up to 0.053 on specificity for Random Forest and 0.052 on MCC for LightGBM), as expected when a single pooled threshold replaces per-fold tuning. The ordering of models is unchanged. The pooled out-of-fold ROC and precision–recall curves (Fig. 7) confirm that all models operate in a high-AUC regime (≈ 0.90), which is expected on a small, near-balanced cohort and implies limited head-room for any single model. The GCN's errors are balanced across types, and its predicted-probability distribution is well separated by class (Fig. 8, right). The calibration curve (Fig. 8, left) indicates reasonable probability calibration after inner-validation threshold tuning.

B. Statistical significance

Table VIII reports the significance tests against *every* baseline in Table VII, not a subset: the two linear/ensemble baselines (Logistic Regression, Random Forest), the three gradient-boosting baselines (Gradient Boosting, LightGBM, XGBoost) and the neural baseline (MLP). None of the twelve tests (six comparisons × two test types) approaches the 0.05 significance level after Holm correction across the family: McNemar's test on the pooled out-of-fold predictions gives $p = 0.70$ against Logistic Regression, $p = 0.77$ against Random Forest, $p = 0.68$ against Gradient Boosting, $p = 0.68$ against MLP, $p = 1.00$ against XGBoost and $p = 0.66$ against LightGBM, and the Wilcoxon signed-rank test on per-fold F1 gives $p = 0.25$, $p = 0.51$, $p = 0.11$, $p = 0.09$, $p = 0.60$ and $p = 0.15$, respectively (Holm-adjusted $p \geq 0.57$ throughout). We therefore characterise the GCN as *statistically indistinguishable from* strong classical baselines—including modern gradient-boosted trees and a comparably-sized MLP—not as a decisive improvement. This honest positioning is, in our view, more credible than an inflated accuracy claim and redirects the contribution toward representation and interpretability.

C. Comparison with the patient-similarity formulation

The premise of this work is that a feature-node graph is a more suitable inductive bias for clinical tabular data than the conventional patient-similarity graph. We test that premise directly.

A k-sweep of the patient-similarity baseline identifies $k = 15$ as its own best setting, which is used throughout. Table IX compares the two formulations under the identical protocol. The feature-node model is ahead on every metric—most clearly on ROC-AUC (0.905±0.041 versus 0.889±0.036)—but the margin is modest. Per-seed DeLong tests [37] on the pooled out-of-fold predictions (Table IX) give $p = 0.351$, 0.910 and 0.017: the direction is consistent across all three seeds, yet the difference reaches significance in only one. *In distribution we therefore claim parity with a consistent directional advantage, not a demonstrated win.*

The more informative difference is structural. The patient-similarity model is transductive: before it can score any patient it requires a cohort-level graph, so it cannot classify an isolated individual. The feature-node model is inductive—each patient is an independent graph—and the topology learned on the development cohort transfers directly. Section VII-L shows that this distinction has practical consequences under distribution shift.

D. Deployment characteristics

Because deployability is one of the three properties on which we rest the case for this formulation (**P2**), we measure it rather than assert it (Table X).

Inductive single-patient inference. We scored held-out patients strictly one at a time, constructing each patient's graph in isolation with no other patient present, and compared the result against batch scoring. The maximum absolute deviation over all patients was 3.0×10^{-8}, i.e. numerically identical: the model's output for a patient does not depend on which other

TABLE VII

PREDICTIVE PERFORMANCE UNDER THE IDENTICAL FIVE-FOLD PROTOCOL. PANEL (A) REPORTS THE FOLD-WISE MEAN ± STANDARD DEVIATION; PANEL (B) REPORTS POOLED OUT-OF-FOLD POINT ESTIMATES OVER ALL 303 PATIENTS.

(a) Fold-wise mean ± std

Model	Accuracy	Recall	F1	ROC-AUC	MCC	Specificity
GCN (Ours)	0.8131 ± 0.0638	0.8011 ± 0.0748	0.7988 ± 0.0595	0.9056 ± 0.0408	0.6320 ± 0.1218	0.8237 ± 0.1170
Logistic Regression	0.8184 ± 0.0524	0.8444 ± 0.0842	0.8100 ± 0.0537	0.9025 ± 0.0385	0.6457 ± 0.1003	0.7971 ± 0.0940
Random Forest	0.7921 ± 0.0558	0.8393 ± 0.0844	0.7893 ± 0.0432	0.9044 ± 0.0336	0.6035 ± 0.0933	0.7524 ± 0.1421
Gradient Boosting	0.7823 ± 0.0496	0.8026 ± 0.0758	0.7716 ± 0.0502	0.8721 ± 0.0390	0.5708 ± 0.0945	0.7648 ± 0.0921
MLP	0.7822 ± 0.0480	0.8227 ± 0.0643	0.7764 ± 0.0450	0.8680 ± 0.0320	0.5728 ± 0.0914	0.7482 ± 0.0849
XGBoost	0.7944 ± 0.0459	0.8515 ± 0.0677	0.7924 ± 0.0386	0.8845 ± 0.0332	0.6027 ± 0.0831	0.7463 ± 0.0981
LightGBM	0.7922 ± 0.0456	0.7914 ± 0.0633	0.7772 ± 0.0493	0.8817 ± 0.0355	0.5844 ± 0.0919	0.7929 ± 0.0527

(b) Pooled out-of-fold point estimates

Model	Accuracy	Recall	F1	ROC-AUC	MCC	Specificity
GCN (Ours)	0.8053	0.7986	0.79	0.9034	0.6087	0.811
Logistic Regression	0.8152	0.8417	0.8069	0.9062	0.6322	0.7927
Random Forest	0.8152	0.8273	0.8042	0.907	0.6303	0.8049
Gradient Boosting	0.7921	0.8201	0.7835	0.8759	0.5864	0.7683
MLP	0.7921	0.8273	0.785	0.8639	0.5876	0.7622
XGBoost	0.802	0.8129	0.7902	0.8825	0.6038	0.7927
LightGBM	0.8185	0.8273	0.807	0.8869	0.6366	0.811

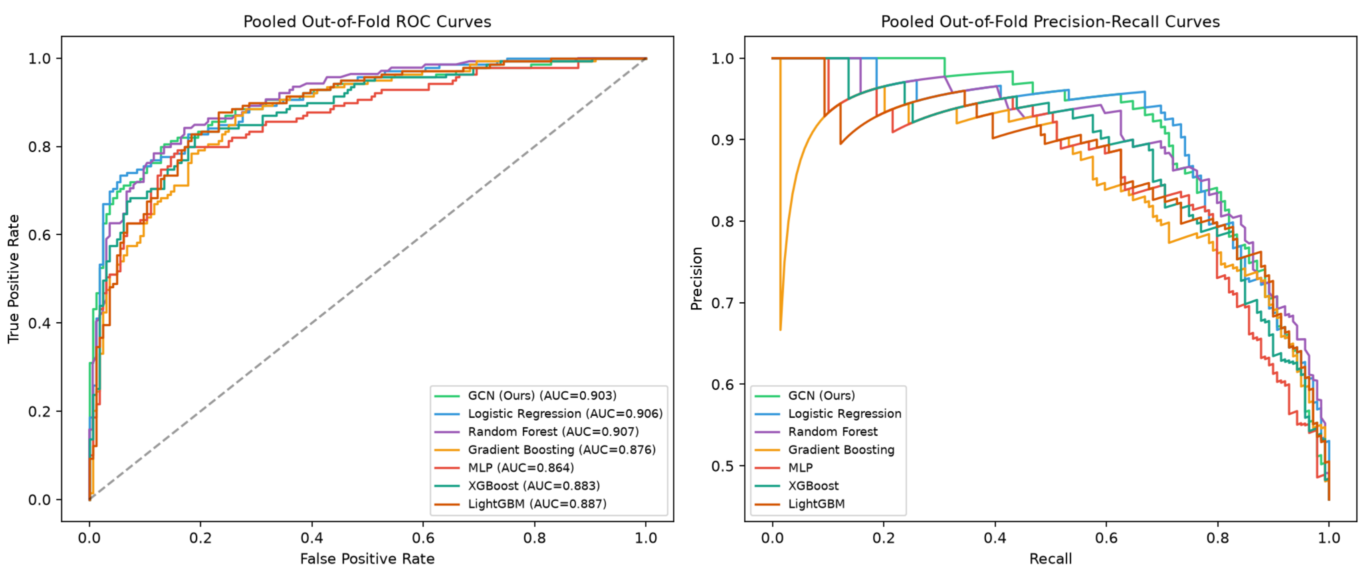

Fig. 7. Pooled out-of-fold ROC (left) and precision–recall (right) curves. All models achieve comparable, high AUC, underscoring the performance ceiling of the cohort.

patients are present. Mean latency was 2.56 ms per patient (median 2.43 ms, p95 2.93 ms) on CPU. The patient-similarity baseline of Section VII-C admits no analogous measurement, because it cannot produce a prediction for an isolated patient at all.

Footprint. The deployed model has 1,281 trainable parameters and serialises to 11.4 KB; training completes in seconds and peak memory stays below 2 MB. Nothing in the pipeline requires a GPU.

Calibration. Discrimination is not sufficient for a decision aid: predicted risks must also be calibrated. Table XI reports Brier score [38] and expected calibration error (ECE) [39]. Out-of-fold on Cleveland the GCN is the best-calibrated of the three models (Brier 0.124, ECE 0.042, against 0.124/0.047 for Logistic Regression and 0.126/0.055 for Random Forest). The more interesting behaviour appears under transport. Logistic Regression's calibration collapses on every external cohort (ECE 0.318, 0.530, 0.471), whereas the GCN degrades far more gracefully on Hungarian (ECE 0.091 versus 0.318 and 0.209)—a 3.5× advantage over the linear baseline—and remains competitive with Random Forest on VA (0.195 versus 0.169). On Switzerland, whose disease prevalence is 93%, all three models are poorly calibrated and Random Forest is the least bad (0.356 versus 0.453). Retaining calibration under distribution shift, not merely ranking ability, is in our view the most practically relevant transport result in this paper.

Clinical utility, and where it is absent. Calibration and discrimination still do not establish that a model is worth

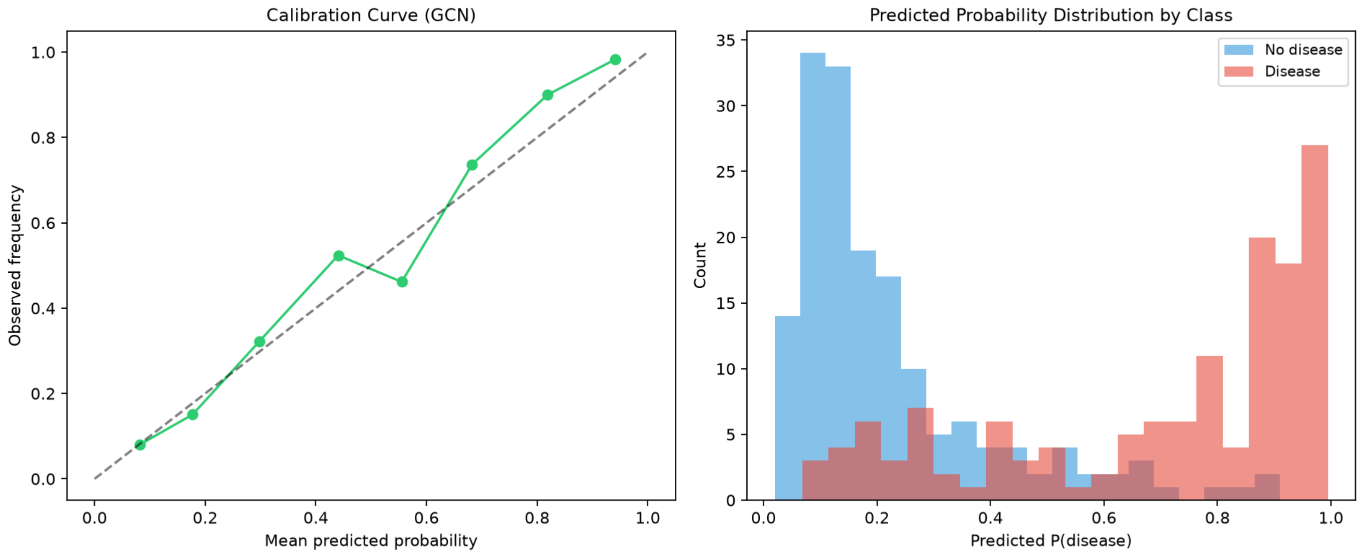

Fig. 8. Left: reliability (calibration) curve for the GCN. Right: distribution of predicted disease probabilities by true class, demonstrating clear separation.

TABLE VIII
STATISTICAL SIGNIFICANCE OF THE GCN VS. BASELINE CLASSIFIERS (MCNEMAR ON POOLED OOF PREDICTIONS; WILCOXON SIGNED-RANK ON PER-FOLD F1).

Comparison	b	c	McNemar stat	McNemar p	Wilcoxon p	Holm p (McNemar)	Sig. Holm (McNemar)	Holm p (Wilcoxon(foldF1))	Sig. Holm (Wilcoxon(foldF1))
GCN vs Logistic Regression	15	12	0.1481	0.7003	0.2455	1.0	no	0.7365	no
GCN vs Random Forest	24	21	0.0889	0.7656	0.5098	1.0	no	1.0	no
GCN vs Gradient Boosting	25	29	0.1667	0.6831	0.107	1.0	no	0.5676	no
GCN vs MLP	24	28	0.1731	0.6774	0.0946	1.0	no	0.5676	no
GCN vs XGBoost	25	26	0.0	1.0	0.5995	1.0	no	1.0	no
GCN vs LightGBM	25	21	0.1957	0.6583	0.1514	1.0	no	0.6056	no

TABLE IX
FEATURE-NODE VERSUS PATIENT-SIMILARITY FORMULATION IN DISTRIBUTION. PANEL (A) REPORTS THE PAIRED COMPARISON; PANEL (B) REPORTS THE SIGNIFICANCE TESTS.

(a) Paired comparison

Formulation	Accuracy	Precision	Recall	F1	ROC-AUC	MCC	Specificity
Feature-node graph (Ours)	0.8109 ± 0.0652	0.8044 ± 0.0999	0.8011 ± 0.0748	0.7971 ± 0.0597	0.9051 ± 0.0413	0.6280 ± 0.1240	0.8196 ± 0.1208
Patient-similarity graph (k=15)	0.8021 ± 0.0467	0.7791 ± 0.0963	0.8275 ± 0.0970	0.7931 ± 0.0447	0.8891 ± 0.0358	0.6184 ± 0.0895	0.7807 ± 0.1163

(b) Significance tests

Seed	AUC (feature-node)	AUC (patient-sim)	ΔAUC	b	c	McNemar p	DeLong p
42.0	0.9033	0.8909	0.0124	22.0	21.0	1.0	0.3506
7.0	0.8915	0.8903	0.0012	20.0	17.0	0.7423	0.9097
123.0	0.9005	0.8726	0.0279	11.0	23.0	0.0592	0.01708

using. We therefore computed decision curves [40], reporting net benefit NB = TP/n − (FP/n) $p_t/(1-p_t)$ across clinically plausible risk thresholds against the default strategy of treating every patient.

The outcome divides cleanly by cohort prevalence, and we report it without selection. On Hungarian, whose 36% prevalence is clinically realistic, the GCN yields the highest net benefit at every threshold examined and exceeds treat-all by a widening margin as p_t grows (0.220 versus 0.085 at $p_t = 0.30$; 0.164 versus −0.281 at $p_t = 0.50$). On Switzerland (93% prevalence) and VA (79%), by contrast, *no* model improves on treating everyone: at $p_t = 0.30$ the GCN attains 0.687 against treat-all's 0.902 on Switzerland, and 0.631 against 0.694 on VA. We regard this as a property of those cohorts rather than of the model—when nearly every patient is diseased, the treat-all strategy is close to optimal by construction and any selective rule can only forgo true positives. It nevertheless bounds the claim we are entitled to make: *demonstrated clinical utility rests on the single external cohort with a realistic prevalence*, and prospective evaluation in a screening population remains necessary.

E. Learned representation

Training is stable under early stopping, and Fig. 9 projects the learned graph-level embeddings with PCA and t-SNE. Disease and non-disease patients form largely separated clusters, indicating that the pooled representation captures class-discriminative structure despite the one-dimensional node features.

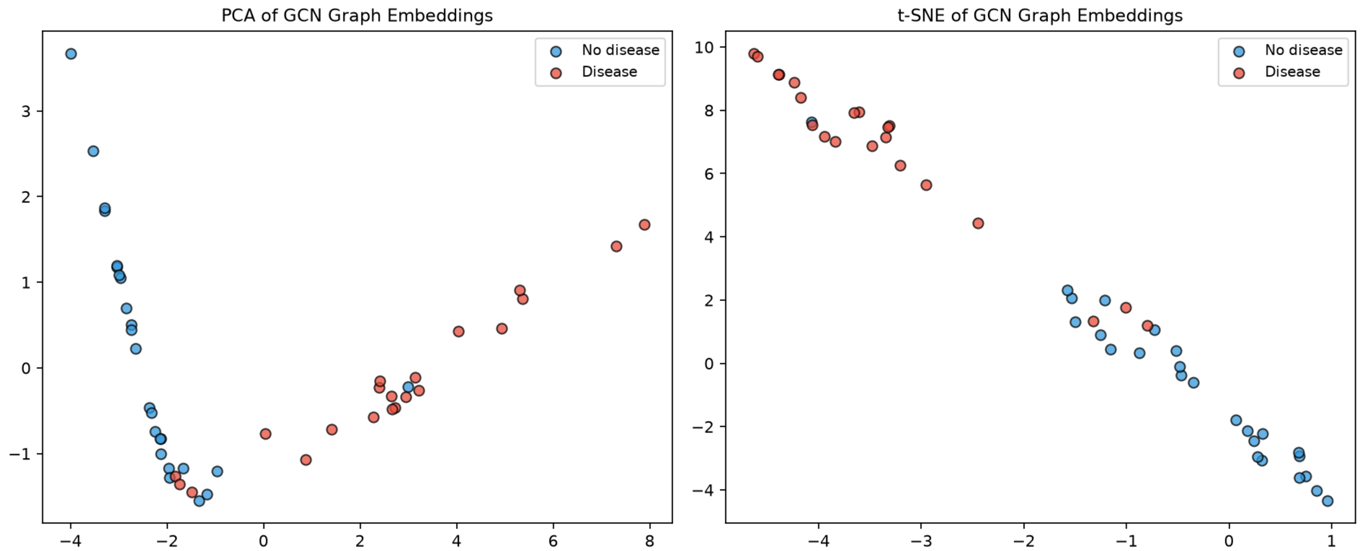

Fig. 9. PCA (left) and t-SNE (right) projections of the learned graph-level embeddings on the held-out test set. The two classes form largely separated clusters.

TABLE X
DEPLOYMENT CHARACTERISTICS OF THE FEATURE-NODE MODEL. INFERENCE IS INDUCTIVE: PATIENTS ARE SCORED INDIVIDUALLY WITH NO COHORT PRESENT.

Property	Value
Scores an isolated patient	Yes (inductive)
Cohort graph required at inference	No
Patients scored individually	46
Max deviation vs batch scoring	2.98e-08
Latency per patient, mean	2.564 ms
Latency per patient, median	2.430 ms
Latency per patient, p95	2.933 ms
Trainable parameters	1,281
Serialised model size	11.4 KB
Hardware	CPU only

F. Ablation study

Table XII presents the complete ablation; panel (b) isolates the effect of each design choice. Four findings emerge.

(i) Selective sparsity helps. The fully connected graph is the weakest configuration (Table XII), confirming that indiscriminate connectivity injects noise and that a sparse, correlation-driven topology is preferable. This is the result that most directly justifies the graph-construction contribution.

(ii) The MST aids connectivity, not accuracy. Removing the MST bridges leaves predictive performance essentially unchanged. This row must be read with care, however: at $\tau = 0.15$ the thresholded graph is already connected in every fold, so the MST adds no edges and the two conditions are in fact the *same graph* (Table III); the residual difference is run-to-run non-determinism rather than a component effect. The augmentation is tested properly in Section IV at $\tau = 0.20$, where it is active: there it restores a single connected component in 5/5 folds without changing any metric significantly. We therefore present the MST as a *structural* guarantee—validated by the positive Fiedler value of Fig. 4—rather than as a source of *internal* accuracy gains. This ablation, like every other row in this table, is computed within the development cohort and is by construction blind to any effect that appears only under distribution shift; Section VII-M examines the same topologies on external cohorts and finds an ordering that this table cannot see.

(iii) Message passing and correlation structure give small, consistent benefits. The correlation graph modestly outperforms an edge-count-matched random graph and the no-edge (independent-node) baseline, indicating that the specific feature-interaction topology, and not merely the presence of edges, carries useful signal.

(iv) Two layers are preferable to one, while the model is robust to readout choice, dropout and hidden width within one standard deviation (Table XII); this robustness is desirable for deployment because it reduces sensitivity to hyperparameter choices.

(v) The choice of message-passing operator is not critical. Because a natural objection to a GCN is that attention-based or more expressive operators might do better, we repeat the entire protocol with GAT, GraphSAGE and GIN substituted for GCNConv, holding graph construction, depth, width, readout, folds and seeds fixed (Table S2, Supplementary Material). No operator differs significantly from GCN: DeLong $p \geq 0.63$ and McNemar $p \geq 0.057$ in every comparison (Table S2). GIN attains a nominally higher accuracy (0.824 versus 0.811), F1 and MCC, but the *lowest* ROC-AUC of the four and requires 2.6× the parameters (3,393 versus 1,281); GAT and GraphSAGE fall between. On a thirteen-node feature graph the neighbourhood structure is evidently too limited for attention or sum-aggregation to exploit. We therefore retain GCN on grounds of *parsimony*—it is the smallest model, attains the

TABLE XI
Calibration of predicted risks. Brier score and expected calibration error (ECE, 10 bins), out-of-fold on Cleveland and on each external cohort. Lower is better.

Setting	Model	Brier	ECE
Cleveland (OOF)	GCN (Ours)	0.1238 ± 0.0023	0.0421 ± 0.0090
Cleveland (OOF)	Logistic Regression	0.1243 ± 0.0033	0.0467 ± 0.0031
Cleveland (OOF)	Random Forest	0.1262 ± 0.0014	0.0546 ± 0.0061
Hungarian	GCN (Ours)	0.1416 ± 0.0059	0.0907 ± 0.0105
Hungarian	Logistic Regression	0.3153 ± 0.0116	0.3181 ± 0.0136
Hungarian	Random Forest	0.1969 ± 0.0069	0.2086 ± 0.0030
Switzerland	GCN (Ours)	0.2975 ± 0.0343	0.4526 ± 0.0340
Switzerland	Logistic Regression	0.5279 ± 0.2047	0.5301 ± 0.2056
Switzerland	Random Forest	0.1944 ± 0.0182	0.3560 ± 0.0220
VA Long Beach	GCN (Ours)	0.2054 ± 0.0085	0.1952 ± 0.0115
VA Long Beach	Logistic Regression	0.4686 ± 0.1730	0.4709 ± 0.1728
VA Long Beach	Random Forest	0.1857 ± 0.0021	0.1693 ± 0.0120

highest ROC-AUC, and is statistically indistinguishable from more expressive alternatives—rather than on grounds of superiority.

The full operator-comparison table (performance and paired significance tests for GAT, GraphSAGE and GIN against GCN) is provided as Table S2 in the Supplementary Material.

G. Threshold sensitivity

Table S1 (Supplementary Material) reports performance as a function of the correlation threshold τ together with the resulting graph density. Performance is robust for $\tau \in [0.10, 0.20]$; $\tau = 0.15$ maximises ROC-AUC while retaining a connected, moderately sparse graph, which motivates our fixed choice. Very dense ($\tau = 0.05$) and very sparse ($\tau = 0.30$) graphs are both inferior, mirroring the topology ablation.

The full per-threshold table is provided as Table S1 in the Supplementary Material.

H. Graph construction rule

The proposed construction retains an edge when $|R_{ij}| \geq \tau$, a single *global* cut-off. A natural alternative is a *per-node* rule: k-nearest-neighbour construction links every feature to its k most strongly correlated partners, fixing a minimum degree of k so that a weakly correlated feature is never left near-isolated by the cut-off. Because this is the most direct objection to a threshold-based design, we test it rather than argue it.

The two families also differ structurally. The global cut-off leaves at least one feature with degree 1 in every fold, whereas k-NN guarantees degree $\geq k$ by construction; $k = 4$ yields almost exactly the same number of edges as our graph (34.6 vs. 34.4), making it a like-for-like control that isolates the *rule* from the *density*. Every k-NN graph is connected in every fold, so the MST is redundant under this rule.

Table XIII reports predictive performance and Table XIII the paired Wilcoxon tests over the 15 matched folds. The Pearson+MST graph attains the highest ROC-AUC (0.9051 vs. 0.8990–0.9036) while the k-NN graphs are nominally stronger on F1 and MCC. Because nine paired tests are performed over the same folds we control the family-wise error rate with the Holm–Bonferroni procedure: *no comparison remains significant after correction* (smallest adjusted $p = 0.163$), and the two nominally significant ROC-AUC differences ($p = 0.018$ and $p = 0.023$ uncorrected) do not survive.

We therefore do not claim that the global cut-off predicts better. The finding is that performance is *insensitive to the construction rule* on this cohort: a global threshold and an adaptive per-node rule of comparable density are statistically indistinguishable. We retain correlation thresholding with MST augmentation because it introduces no hyper-parameter beyond τ, because its edge set is directly readable as a correlation structure, and because it admits the spectral interpretation used in Section IV—not because it is more accurate.

I. What message passing over scalar nodes contributes

Each node carries a single scalar measurement, so it is fair to ask what propagation can add. Mechanically the answer is clear. The first graph convolution maps $\mathbb{R}^1 \to \mathbb{R}^d$ per node and then averages over neighbours, so after one layer a feature's representation is no longer a function of its own measurement alone but of the measurements of the features it is correlated with. Maximum heart rate, for instance, is joined in *every* fold to chest-pain type, exercise-induced angina, ST depression, slope, number of coloured vessels and thalassaemia; its embedding is therefore formed in the context of the exercise-ischaemia cluster rather than in isolation. Cholesterol, by contrast, retains only age as a fold-stable neighbour, and is contextualised correspondingly little.

We are deliberately precise about how much this buys, because the scalar node dimension bounds it. Since the input dimension is one, the layer-1 pre-activation of every node is a scalar multiple of the same weight vector: the node representation is *rank-one in direction by construction*. Measured on the trained model, the participation-ratio effective rank of the 13×32 embedding matrix rises only to 1.63 after the first layer and 1.73 after the second, against a possible 13. The learned representation is thus close to a scalar score per feature, and the pooled graph-level function is correspondingly close to a linear function of graph-smoothed inputs. We regard this as explanatory rather than embarrassing: it is the mechanistic counterpart of the accuracy parity with logistic regression reported in Section VII, and it makes that parity an expected consequence of the design rather than a surprise. It is also why depth does not help here—propagation reduces the effective

TABLE XII
ABLATION STUDY. PANEL (A) REPORTS ABSOLUTE PERFORMANCE FOR EACH VARIANT, POOLED OVER 3 SEEDS × 5 FOLDS; PANEL (B) REPORTS THE CHANGE RELATIVE TO THE FULL MODEL.

(a) Absolute performance

Experiment	Accuracy	F1	ROC-AUC	MCC	Recall	Specificity
Full (Corr+MST)	0.8109 ± 0.0652	0.7971 ± 0.0597	0.9051 ± 0.0413	0.6280 ± 0.1240	0.8011 ± 0.0748	0.8196 ± 0.1208
w/o MST (Corr only)	0.8032 ± 0.0787	0.7936 ± 0.0692	0.9027 ± 0.0437	0.6164 ± 0.1468	0.8105 ± 0.0825	0.7974 ± 0.1464
Random graph	0.7955 ± 0.0614	0.7896 ± 0.0545	0.8959 ± 0.0414	0.6056 ± 0.1115	0.8300 ± 0.0892	0.7669 ± 0.1303
Fully connected	0.8021 ± 0.0753	0.7920 ± 0.0729	0.8973 ± 0.0432	0.6107 ± 0.1428	0.8153 ± 0.0869	0.7911 ± 0.1208
No graph (indep. nodes)	0.8086 ± 0.0661	0.7916 ± 0.0685	0.8966 ± 0.0485	0.6262 ± 0.1276	0.7942 ± 0.1073	0.8214 ± 0.1191
1 GCN layer	0.8064 ± 0.0606	0.7906 ± 0.0628	0.9005 ± 0.0423	0.6187 ± 0.1195	0.7963 ± 0.0916	0.8155 ± 0.1052
Max pooling	0.8032 ± 0.0778	0.7979 ± 0.0687	0.8940 ± 0.0518	0.6179 ± 0.1427	0.8347 ± 0.0827	0.7770 ± 0.1389
Add pooling	0.8042 ± 0.0644	0.7936 ± 0.0566	0.9008 ± 0.0361	0.6200 ± 0.1198	0.8132 ± 0.0900	0.7973 ± 0.1331
Dropout 0.0	0.8218 ± 0.0612	0.8038 ± 0.0586	0.9040 ± 0.0410	0.6473 ± 0.1189	0.7914 ± 0.0762	0.8480 ± 0.1037
Dropout 0.5	0.8130 ± 0.0641	0.7988 ± 0.0599	0.9061 ± 0.0408	0.6319 ± 0.1223	0.8011 ± 0.0748	0.8237 ± 0.1170
Hidden 16	0.7660 ± 0.1328	0.7752 ± 0.0773	0.7993 ± 0.2716	0.5491 ± 0.2413	0.8250 ± 0.1046	0.7166 ± 0.3005
Hidden 64	0.8163 ± 0.0587	0.8009 ± 0.0586	0.9040 ± 0.0414	0.6355 ± 0.1136	0.8034 ± 0.0782	0.8272 ± 0.0939

(b) Change relative to the full model

Experiment	Accuracy	F1	ROC-AUC	MCC	Recall	Specificity
w/o MST (Corr only)	-0.0077	-0.0035	-0.0023	-0.0116	0.0094	-0.0222
Random graph	-0.0153	-0.0075	-0.0092	-0.0223	0.0289	-0.0527
Fully connected	-0.0088	-0.0051	-0.0077	-0.0173	0.0142	-0.0285
No graph (indep. nodes)	-0.0023	-0.0055	-0.0085	-0.0018	-0.0069	0.0018
1 GCN layer	-0.0044	-0.0065	-0.0045	-0.0092	-0.0048	-0.0041
Max pooling	-0.0077	0.0009	-0.011	-0.0101	0.0336	-0.0426
Add pooling	-0.0066	-0.0035	-0.0042	-0.008	0.0122	-0.0223
Dropout 0.0	0.0109	0.0068	-0.001	0.0194	-0.0096	0.0283
Dropout 0.5	0.0022	0.0017	0.001	0.0039	0.0	0.004
Hidden 16	-0.0449	-0.0219	-0.1058	-0.0789	0.024	-0.103
Hidden 64	0.0054	0.0038	-0.001	0.0076	0.0024	0.0076

rank relative to processing the features independently (1.73 versus 2.22), which is smoothing, not enrichment.

The ablation supports this reading and no stronger reading. Under Holm correction over nine paired tests (Table XV), replacing the correlation graph with a *random* graph of identical density significantly reduces ROC-AUC (0.9051 → 0.8959, adjusted $p = 0.005$): when propagation does occur, it matters that the edges encode genuine clinical correlation rather than arbitrary adjacency. But neither removing the graph entirely (0.8966, adjusted $p = 0.249$) nor making it fully connected (0.8973, adjusted $p = 0.205$) differs significantly from our topology, and exchanging GCN for GAT, GraphSAGE or GIN changes no metric significantly either (Section VII-F). We therefore claim exactly what the evidence supports: *the choice of topology matters more than the choice of operator, and a correlation-derived topology is measurably better than an arbitrary one*—while explicitly not claiming that message passing over scalar nodes outperforms treating the features independently. The value of the formulation on this cohort lies in the structure it exposes and the explanations it supports, not in a predictive gain from propagation.

A falsification test. The account above attributes the graph's contribution to *smoothing* along correlated features rather than to relational reasoning over them. That distinction is testable. Relational reasoning requires the network to distinguish one feature from another, which the scalar encoding cannot do: with shared weights and permutation-equivariant aggregation, two features with similar values and similar degree are indistinguishable. Supplying that missing information should therefore help if the graph reasons relationally, and should interfere if it smooths. We repeated the full protocol with a learnable per-feature identity embedding concatenated to each node's scalar value ($d = 8$), holding graph construction, depth, width, readout, folds and seeds fixed (Table XIV).

The embedding behaves exactly as the smoothing account predicts. Without edges it does what it was designed to do: effective rank rises from 1.01 to 2.78 and ROC-AUC improves to 0.9121. With edges present the gain is undone—rank falls back to 1.63 for Corr+MST and to 1.00 for the fully connected graph, and ROC-AUC drops to 0.8237 and 0.8347 respectively. The ordering is monotone in edge density: the more the graph mixes nodes, the less identity survives. Consequently the graph gap reverses sign under the identity encoding (Table XIV): +0.0086 with scalar nodes (11/15 paired wins, $p = 0.031$) becomes −0.0884 with identity embeddings. Message passing over feature nodes and explicit feature identity are in direct tension, and the formulation works because the node representation is a bare scalar with no identity to destroy. We report this as a negative result that constrains the interpretation: it rules out the relational reading that the feature-node framing might otherwise invite, and it identifies richer node features as a direction that is *not* straightforwardly beneficial under this operator.

Does the reversal generalise beyond $d = 8$? The single dimension tested above could have been a coincidence of that particular embedding size. We repeated the identity-embedding experiment at $d \in \{0, 2, 4, 8, 16, 32\}$, holding everything else fixed (Table S3, Supplementary Material). The sign reversal

TABLE XIII
k-NEAREST-NEIGHBOUR FEATURE TOPOLOGY. PANEL (A) REPORTS PERFORMANCE ACROSS k; PANEL (B) REPORTS THE PAIRED TESTS AGAINST THE PEARSON+MST GRAPH.

(a) Performance across k

Construction	Accuracy	F1	ROC-AUC	MCC	Recall	Specificity
Corr+MST, τ=0.15 (Ours)	0.8109 ± 0.0652	0.7971 ± 0.0597	0.9051 ± 0.0413	0.6280 ± 0.1240	0.8011 ± 0.0748	0.8196 ± 0.1208
Corr only, τ=0.15	0.8032 ± 0.0787	0.7936 ± 0.0692	0.9027 ± 0.0437	0.6164 ± 0.1468	0.8105 ± 0.0825	0.7974 ± 0.1464
k-NN, k=2	0.8185 ± 0.0608	0.8076 ± 0.0585	0.8990 ± 0.0429	0.6429 ± 0.1195	0.8249 ± 0.0751	0.8134 ± 0.1032
k-NN, k=3	0.8054 ± 0.1102	0.8021 ± 0.0756	0.9015 ± 0.0406	0.6180 ± 0.2033	0.8200 ± 0.0819	0.7932 ± 0.2300
k-NN, k=4	0.8163 ± 0.0632	0.8023 ± 0.0632	0.9036 ± 0.0402	0.6363 ± 0.1209	0.8102 ± 0.0805	0.8212 ± 0.1022

(b) Paired tests vs. Pearson+MST

Comparison	Metric	Mean A	Mean B	Δ (A−B)	Wilcoxon p	Holm p	Significant (Holm)
Corr+MST, τ=0.15 (Ours) vs k-NN, k=2	F1	0.7971	0.8076	-0.0105	0.1981	1.0	no
Corr+MST, τ=0.15 (Ours) vs k-NN, k=2	ROC-AUC	0.9051	0.899	0.0061	0.0181	0.1629	no
Corr+MST, τ=0.15 (Ours) vs k-NN, k=2	MCC	0.628	0.6429	-0.0149	0.5098	1.0	no
Corr+MST, τ=0.15 (Ours) vs k-NN, k=3	F1	0.7971	0.8021	-0.005	0.2209	1.0	no
Corr+MST, τ=0.15 (Ours) vs k-NN, k=3	ROC-AUC	0.9051	0.9015	0.0036	0.023	0.184	no
Corr+MST, τ=0.15 (Ours) vs k-NN, k=3	MCC	0.628	0.618	0.01	0.5098	1.0	no
Corr+MST, τ=0.15 (Ours) vs k-NN, k=4	F1	0.7971	0.8023	-0.0052	0.6496	1.0	no
Corr+MST, τ=0.15 (Ours) vs k-NN, k=4	ROC-AUC	0.9051	0.9036	0.0015	0.4698	1.0	no
Corr+MST, τ=0.15 (Ours) vs k-NN, k=4	MCC	0.628	0.6363	-0.0083	0.9721	1.0	no

of the graph gap holds at *every* tested dimension, not only d = 8: ΔROC-AUC (Corr+MST − no graph) is +0.0089 at d = 0 and becomes negative at d = 2 (−0.0060), d = 4 (−0.0081), d = 8 (−0.0882), d = 16 (−0.0095) and d = 32 (−0.0605) (Table S4, Supplementary Material); none of the twelve (dimension, contrast) pairs survives Holm correction, consistent with the modest effect sizes already reported at d = 8. The magnitude is not monotone in d: the reversal is small at d = 2, d = 4 and d = 16 but sharply larger at d = 8 and d = 32, where the Corr+MST and fully connected conditions also become numerically unstable (standard deviation of ROC-AUC rising to 0.18–0.29, versus 0.04 at d = 0). The effective rank confirms the same smoothing mechanism throughout the range: with no graph, rank rises monotonically with d (from 1.01 at d = 0 to 5.57 at d = 32), whereas the fully connected graph collapses identity back to rank 1.00 at *every* tested dimension and Corr+MST sits in between, always closer to the fully connected value than to the no-graph value. The reversal is therefore not an artefact of one embedding size; it is a property of message passing over identity-augmented scalar nodes at any dimension we tested, which strengthens the mechanistic account above rather than merely illustrating it at a single point.

The full dimension-sweep tables (performance/effective-rank at each d, and the corresponding paired graph-gap tests) are provided as Tables S3 and S4 in the Supplementary Material.

J. Explainability

Table XVI compares three attribution methods across eight metrics. Integrated Gradients yields the most faithful and sparsest explanations—leading on fidelity, sparsity and both deletion and insertion area (Fig. 10)—while Saliency is the most stable but least faithful. The mean feature-importance profiles concentrate on `cp`, `oldpeak`, `thalach`, `ca` and `thal`. We call this *literature-based clinical agreement*: the comparison is against risk factors reported in prior cardiovascular studies [25], [2], not against a blinded clinician panel scoring our specific explanations, and should be read with that scope in mind.

Crucially, GNNExplainer and Integrated Gradients agree strongly (Table XVI; Spearman ρ = 0.80, top-k Jaccard 0.59). Because these two mechanistically different explainers converge on the same risk features, their agreement provides evidence of explanation reliability that a single method cannot offer. We therefore recommend GNNExplainer as the primary, graph-native explainer, corroborated by Integrated Gradients as an axiomatic cross-check. Saliency, whose raw gradients are noisier, is reported for completeness.

K. Error analysis

Fig. 11 shows that misclassifications concentrate near the decision boundary: the model is rarely confidently wrong, which is desirable in a screening context where borderline cases can be flagged for clinician review.

L. Cross-site transport within the UCI heart-disease collection

Terminology and scope. We refer to the Hungarian, Switzerland and Long-Beach-VA cohorts as *external* to the Cleveland cohort on which every component is fit, and to this evaluation as external validation, in the narrow sense that no observation from these three cohorts is used at any stage of model development. We do not intend the stronger sense in which the term is sometimes used—validation on a cohort collected by an unrelated group, under an unrelated protocol, in an unrelated era. All four cohorts are constituent databases of the same UCI Heart Disease collection [24], [25], assembled by coordinated investigators using a shared variable definition and recruitment protocol in the late 1980s. Site, and to some degree the recruiting era, therefore differ across cohorts, but the broader clinical and measurement lineage does not, and the transport result below should be read as cross-site evaluation within one historical data collection rather than validation against an independent one.

TABLE XIV
NODE-IDENTITY FALSIFICATION TEST. PANEL (A) REPORTS PERFORMANCE AND EFFECTIVE RANK UNDER EACH NODE ENCODING; PANEL (B) REPORTS THE INTERNAL GRAPH GAP PAIRED OVER THE 15 MATCHED RUNS.

(a) Performance and effective rank

Node encoding	Topology	Effective rank (conv1)	ROC-AUC	MCC
Scalar node (current)	No graph	1.01	0.8966 ± 0.0485	0.6262 ± 0.1276
Scalar node (current)	Corr+MST	1.0	0.9051 ± 0.0413	0.6280 ± 0.1240
Scalar node (current)	Fully connected	1.0	0.8973 ± 0.0432	0.6107 ± 0.1428
+ Identity embedding (d=8)	No graph	2.78	0.9121 ± 0.0399	0.6623 ± 0.1120
+ Identity embedding (d=8)	Corr+MST	1.63	0.8237 ± 0.2296	0.5561 ± 0.2482
+ Identity embedding (d=8)	Fully connected	1.0	0.8347 ± 0.1991	0.5305 ± 0.2459

(b) Paired internal graph gap

Node encoding	Contrast	ΔROC-AUC	Wilcoxon p (15 paired folds)	Wins / 15
Scalar node (current)	Corr+MST − No graph	0.0086	0.0309	11
Scalar node (current)	Corr+MST − Fully connected	0.0078	0.0256	11
+ Identity embedding (d=8)	Corr+MST − No graph	-0.0884	0.0833	6
+ Identity embedding (d=8)	Corr+MST − Fully connected	-0.011	0.4431	10

TABLE XV
TOPOLOGY SIGNIFICANCE. PANEL (A) REPORTS PERFORMANCE BY TOPOLOGY; PANEL (B) REPORTS THE PAIRED CONTRASTS WITH MULTIPLICITY CONTROL.

(a) Performance by topology

Topology	Accuracy	F1	ROC-AUC	MCC
Corr+MST (Ours)	0.8109 ± 0.0652	0.7971 ± 0.0597	0.9051 ± 0.0413	0.6280 ± 0.1240
No graph	0.8086 ± 0.0661	0.7916 ± 0.0685	0.8966 ± 0.0485	0.6262 ± 0.1276
Fully connected	0.8021 ± 0.0753	0.7920 ± 0.0729	0.8973 ± 0.0432	0.6107 ± 0.1428
Random graph	0.7955 ± 0.0614	0.7896 ± 0.0545	0.8959 ± 0.0414	0.6056 ± 0.1115

(b) Paired contrasts

Comparison	Metric	Ours	Other	Δ (Ours−Other)	Wilcoxon p	Holm p	Significant (Holm)
Corr+MST (Ours) vs No graph	F1	0.7971	0.7916	0.0055	0.6292	1.0	no
Corr+MST (Ours) vs No graph	ROC-AUC	0.9051	0.8966	0.0085	0.0356	0.2492	no
Corr+MST (Ours) vs No graph	MCC	0.628	0.6262	0.0018	0.4887	1.0	no
Corr+MST (Ours) vs Fully connected	F1	0.7971	0.792	0.0051	0.8904	1.0	no
Corr+MST (Ours) vs Fully connected	ROC-AUC	0.9051	0.8973	0.0077	0.0256	0.2048	no
Corr+MST (Ours) vs Fully connected	MCC	0.628	0.6107	0.0173	0.5995	1.0	no
Corr+MST (Ours) vs Random graph	F1	0.7971	0.7896	0.0075	0.6002	1.0	no
Corr+MST (Ours) vs Random graph	ROC-AUC	0.9051	0.8959	0.0092	0.0006	0.0054	yes
Corr+MST (Ours) vs Random graph	MCC	0.628	0.6056	0.0223	0.1961	1.0	no

Feature availability constrains what can be validated. The primary 13-feature model cannot be transported: the non-Cleveland UCI cohorts systematically do not record `ca` (95–99% missing), `thal` (42–90%) or `slope` (14–65%), and the Switzerland `chol` column is entirely zero-encoded, a documented sentinel rather than a measurement. Complete-case counts for the full feature vector are 1/294 (Hungarian), 0/123 (Switzerland) and 1/200 (VA), so cross-cohort transport evaluation of the full model is impossible in principle, not merely inconvenient (Table XVIII). We therefore define a *transportable* subset of the eight variables recorded in all four cohorts—`age`, `sex`, `cp`, `trestbps`, `restecg`, `thalach`, `exang` and `oldpeak`—and retrain the identical architecture on Cleveland restricted to those features. Discarding `chol` and `fbs` costs little (target correlations +0.085 and +0.025), whereas discarding `ca` (+0.460), `thal` (+0.516) and `slope` (+0.339) is a genuine loss forced by data availability rather than chosen. That cost is visible internally: cross-validated ROC-AUC on Cleveland falls from 0.905 ± 0.041 (13 features) to 0.840 ± 0.038 (8 features, Table XVIII), which separates the price of feature reduction from the price of cohort shift.

Protocol. The scaler, the correlation+MST graph, the network weights and the decision threshold are all fit on Cleveland alone; each external cohort is scored exactly once, baselines are transported under the identical protocol, and the procedure is repeated over five seeds. The transport, calibration and decision-curve analyses that follow compare three models—the feature-node GCN, Logistic Regression and Random Forest—rather than all seven of Table VII: these are the strongest linear and ensemble baselines in distribution, and holding the transported set to three keeps the five-seed × three-cohort protocol and its bootstrap intervals interpretable. The boosted-tree baselines are reported in distribution only.

Result. The feature-node GCN attains a higher ROC-AUC than both baselines on all three external cohorts (Hungarian 0.876 ± 0.004, Switzerland 0.744 ± 0.008, VA 0.721 ± 0.007), so the sign of the difference is positive in all six model–cohort comparisons (Table XIX, Fig. 12). On Hungarian the transported AUC (0.876) exceeds the model's own internal Cleveland AUC on the same eight features (0.840); we note this but do not read it as superior transfer, since ROC-AUC is not comparable across cohorts of different case mix. This contrasts with the in-distribution result of Section VII, where the GCN was statistically indistinguishable from the same

TABLE XVI
EXPLAINABILITY EVALUATION. PANEL (A) COMPARES THE THREE ATTRIBUTION METHODS ACROSS THE QUANTITATIVE METRICS; PANEL (B) REPORTS CROSS-METHOD AGREEMENT.

(a) Attribution-method metrics

Metric	GNNExplainer	Integrated Gradients	Saliency
Fidelity+	0.4982	0.5181	0.4421
Fidelity-	0.7188	0.7953	0.5951
Sparsity	0.4731	0.7404	0.3462
Stability	0.9605	0.9991	0.9997
Sensitivity(↓)	0.0395	0.0009	0.0003
Deletion-AUC(↓)	0.0971	0.0767	0.1231
Insertion-AUC	0.3389	0.3705	0.2943
Clinical-Agreement	0.505	0.615	0.75

(b) Cross-method agreement

Method pair	Top-k Jaccard	Spearman rank corr
GNNExplainer vs Integrated Gradients	0.5877	0.8046
GNNExplainer vs Saliency	0.2808	0.2593
Integrated Gradients vs Saliency	0.3457	0.3639

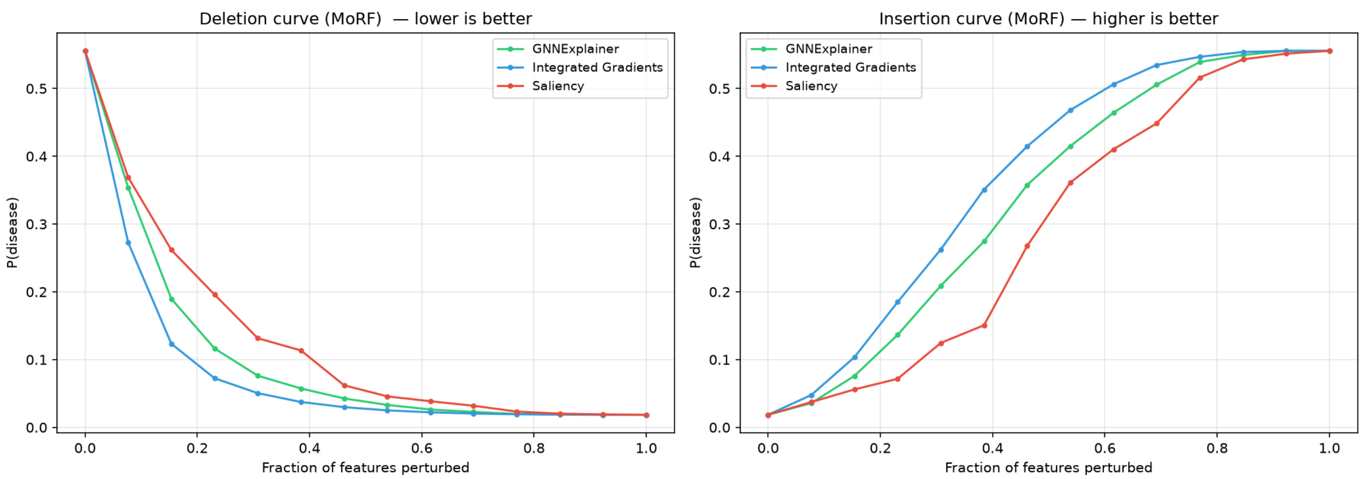

Fig. 10. Deletion (left, lower is better) and insertion (right, higher is better) curves under most-relevant-first perturbation. Integrated Gradients produces the sharpest drop and fastest recovery.

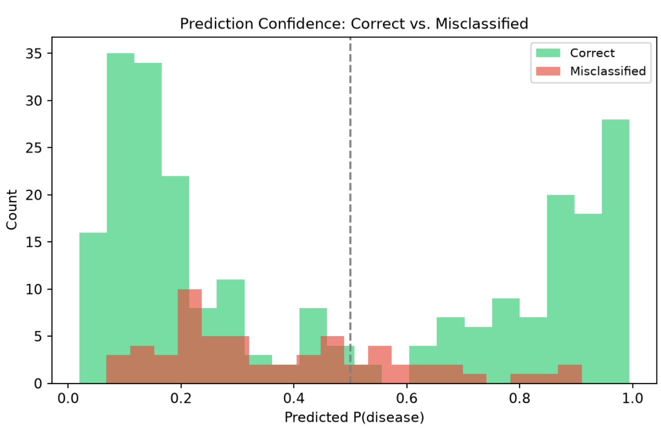

Fig. 11. Error analysis. Correct predictions are confident, whereas misclassifications cluster around $P = 0.5$, indicating calibrated uncertainty rather than confident errors.

baselines, and suggests that the graph prior buys robustness to covariate and prevalence shift rather than in-distribution accuracy: restricting message passing to statistically related features appears to encode structure that is more stable across populations than the marginal decision surfaces the tabular baselines learn. Random Forest in particular degenerates toward majority-class prediction under shift, its specificity falling to 0.39, 0.20 and 0.11 across the three cohorts, whereas the GCN retains a balanced operating point; Logistic Regression retains comparable specificity but at substantially lower AUC.

The patient-similarity formulation transports less reliably. We transported the patient-similarity GNN of Section VII-C under the identical protocol, with k fixed at the value selected on Cleveland and never re-tuned (Table XVII). The feature-node model attains the higher ROC-AUC on two of three cohorts—Hungarian (+0.011) and VA (+0.055)—while Switzerland marginally favours the patient-similarity model (−0.009, Table XVII); we report that split result as it stands. The difference in *failure mode* is nonetheless stark: on VA the patient-similarity model degenerated to a single-class predictor in all five seeds, labelling every patient diseased (recall 1.000 ± 0.000, specificity 0.000 ± 0.000, MCC $0.000 \pm$

0.000), and approached the same degeneracy on Switzerland (specificity 0.150), whereas the feature-node model retained a usable operating point on both. Because the transductive model must rebuild its neighbourhood graph from the external cohort itself, a shifted covariate distribution reshapes the very structure over which messages pass; the feature-node model instead re-uses a fixed, Cleveland-derived feature topology unchanged.

Statistical support, honestly stated. DeLong's test on the correlated ROC curves spans six comparisons (three cohorts × two baselines), so we judge it on Holm-adjusted p-values (Table XIX). Only one survives correction: Logistic Regression on Hungarian (ΔAUC = +0.169, raw $p = 5 \times 10^{-8}$, adjusted $p < 10^{-4}$). The two comparisons nominally significant before correction—VA against Logistic Regression (+0.162, raw $p = 0.022$) and Switzerland against Random Forest (+0.126, raw $p = 0.020$)—do not survive (both adjusted $p = 0.100$). These p-values must be read alongside the sampling uncertainty, which is severe on two cohorts: Switzerland contributes only eight negative cases and VA thirty, so a seed-to-seed standard deviation, which measures only initialisation variability, drastically understates the true uncertainty. We therefore computed a nested bootstrap ($B = 2000$) resampling patients in a class-stratified manner *and* drawing over the five fitted models (Table XX). The effect is large: the Switzerland AUC is 0.745 with a 95% interval of $[0.543, 0.927]$—a width of 0.38 against a seed-only spread of 0.008—so any conclusion from Switzerland alone is weakly supported irrespective of the point estimate. Paired bootstrap intervals for ΔAUC (Table XX) give the most conservative summary: the advantage over Logistic Regression is established on Hungarian (+0.169, $[+0.108, +0.231]$) and VA (+0.151, $[+0.020, +0.289]$), while every remaining comparison, including all three against Random Forest, has an interval containing zero. The Switzerland contrast against Random Forest, which DeLong marks significant, has a paired interval of $[-0.091, +0.215]$; with eight negative cases we regard the bootstrap as more reliable and do not claim that advantage. We therefore state the external finding as follows: the feature-node GCN is *directionally superior in all six model–cohort comparisons*, its advantage over Logistic Regression is *established on two of three cohorts*, and its advantage over Random Forest is *not established on any*.

Caveats. Prevalence shifts from 46% to 93% across cohorts, so threshold-dependent metrics are hard to interpret; Hungarian, with 292 complete cases and a 36% prevalence closest to Cleveland's, is the most trustworthy of the three and is also where the advantage is clearest. Complete-case filtering is not neutral for VA: 60 of 200 records (30.0%) are dropped and prevalence among retained cases (78.6%) is 4.1 percentage points higher than in the full cohort (74.5%), indicating mild outcome-dependent missingness, whereas Hungarian and Switzerland show no comparable shift (< 0.5 pp, Table XVIII). No imputation is used at any stage, so this selection effect, not an imputation artefact, is the source of any residual bias on VA.

M. τ-regime analysis of graph connectivity

The internal ablation found no benefit from the graph, and at our operating point the MST is inert; both could be read as evidence that the topology is decorative. We tested an alternative reading—that its contribution is transportability rather than in-distribution accuracy, and is therefore invisible to any internal experiment. Whether the MST acts at all depends jointly on the threshold and the feature set: the 13-feature graph stays connected for $\tau \leq 0.15$ and fragments from $\tau = 0.20$, whereas the reduced 8-feature transportable set already fragments at $\tau = 0.15$. This retrospectively explains the null "w/o MST" ablation row, because at $\tau = 0.15$ on 13 features the two conditions describe the identical graph. Repeating the transport protocol of Section VII-L at $\tau = 0.20$, where the MST is active, the internal and external orderings dissociate: internally the ranking is flat and slightly favours the edgeless model (validation ROC-AUC 0.8293 against 0.8276), whereas externally Corr+MST attains the highest ROC-AUC on all three cohorts and the edgeless model falls to 0.6343 on Switzerland against 0.7433. Page's trend test for the ordered alternative is significant pooled ($L = 406$, $p = 1.4 \times 10^{-9}$ over 30 seed×cohort blocks) [41], and a fully connected graph, which is also a single component and therefore isolates selectivity from connectivity, is significantly worse than Corr+MST on all three cohorts (ΔAUC +0.0117 to +0.0572, all $p \leq 0.006$). What transports is thus a *sparse, correlation-selective* topology that is also connected—not density, and not connectivity on its own. We bound the claim to what the evidence supports: these tests are computed over random initialisations, and under paired patient-level bootstrap only 3 of 9 contrasts exclude zero, none surviving Holm correction of the DeLong tests, with modest effect sizes (ΔAUC +0.005 to +0.057). We therefore describe this only as *suggestive evidence* that a sparse, connected topology may be more favourable under distribution shift, consistent in direction across three cohorts and reproducibly across initialisations, with patient-level confidence established on the best-powered cohort alone. We do not claim that the MST improves prediction; internally it does not, and we retain it at $\tau = 0.15$ as a connectivity safeguard, as stated in Section IV-B. The complete tables for this analysis accompany the released code.

VIII. Discussion

We organise the discussion around the three properties set out in Section I, then state what the evidence does not support.

P1: Representation. The feature-node graph turns an implicit modelling assumption into an explicit, inspectable object: the topology is available for spectral and centrality analysis *before* any training occurs (Section IV), so the structure a clinician is asked to trust can be examined independently of model fit—a property no tabular baseline offers. Downstream, three mechanistically distinct attribution methods converge on similar rankings (GNNExplainer and Integrated Gradients: Spearman $\rho = 0.80$, top-k Jaccard 0.59) that align with established cardiovascular risk factors. Convergence across methods matters more than any single attribution: it indicates the explanation reflects the learned function rather than an

TABLE XVII
TRANSPORT OF THE PATIENT-SIMILARITY FORMULATION. PANEL (A) REPORTS ITS EXTERNAL PERFORMANCE; PANEL (B) CONTRASTS IT WITH THE FEATURE-NODE MODEL PER COHORT.

(a) Patient-similarity, transported

Cohort	Model	n	Accuracy	Precision	Recall	F1	ROC-AUC	MCC	Specificity
Hungarian	Patient-similarity GNN (k=15)	292	0.7486 ± 0.0749	0.6312 ± 0.0911	0.8343 ± 0.0891	0.7093 ± 0.0493	0.8648 ± 0.0175	0.5277 ± 0.0893	0.7005 ± 0.1501
Switzerland	Patient-similarity GNN (k=15)	116	0.9103 ± 0.0372	0.9399 ± 0.0145	0.9667 ± 0.0579	0.9517 ± 0.0227	0.7537 ± 0.0693	0.0968 ± 0.1269	0.1500 ± 0.2424
VA Long Beach	Patient-similarity GNN (k=15)	140	0.7857 ± 0.0000	0.7857 ± 0.0000	1.0000 ± 0.0000	0.8800 ± 0.0000	0.6662 ± 0.0397	0.0000 ± 0.0000	0.0000 ± 0.0000

(b) Contrast per cohort

Cohort	AUC feature-node	AUC patient-sim	ΔAUC
Hungarian	0.8755	0.8648	0.0107
Switzerland	0.7444	0.7537	-0.0093
VA Long Beach	0.7211	0.6662	0.0549

TABLE XVIII
EXTERNAL COHORT SETUP. PANEL (A) REPORTS COHORT AVAILABILITY UNDER THE 8-FEATURE TRANSPORTABLE SET; PANEL (B) GIVES THE INTERNAL CLEVELAND REFERENCE FOR THAT REDUCED MODEL, SEPARATING THE COST OF FEATURE REDUCTION FROM THE COST OF COHORT SHIFT.

(a) Cohort availability

Cohort	Total records	Complete cases (8 features)	Missing (dropped)	Missing %	Prevalence (all records)	Prevalence (complete cases)	Positive (n)	Negative (n)	Status
Hungarian	294	292	2	0.7	36.1%	36.0%	105	187	evaluated
Switzerland	123	116	7	5.7	93.5%	93.1%	108	8	evaluated
VA Long Beach	200	140	60	30.0	74.5%	78.6%	110	30	evaluated

(b) Internal 8-feature reference

Setting	Accuracy	Recall	F1	ROC-AUC	MCC	Specificity
Cleveland internal CV (8-feature)	0.7275 ± 0.0783	0.8041 ± 0.0996	0.7333 ± 0.0463	0.8400 ± 0.0379	0.4819 ± 0.1210	0.6622 ± 0.2007

TABLE XIX
EXTERNAL VALIDATION RESULTS. PANEL (A) REPORTS TRANSPORTED PERFORMANCE PER COHORT (MEAN ± STD OVER 5 SEEDS); PANEL (B) REPORTS THE CORRESPONDING DELONG AND MCNEMAR COMPARISONS WITH HOLM CORRECTION.

(a) Transported performance

Cohort	Model	n	Recall	F1	ROC-AUC	MCC	Specificity
Hungarian	GCN (Ours)	292	0.6629 ± 0.0712	0.7080 ± 0.0195	0.8755 ± 0.0037	0.5715 ± 0.0075	0.8845 ± 0.0456
Hungarian	Logistic Regression	292	0.3810 ± 0.1927	0.4361 ± 0.1069	0.7060 ± 0.0025	0.2817 ± 0.0324	0.8492 ± 0.1262
Hungarian	Random Forest	292	0.9219 ± 0.0904	0.6201 ± 0.0461	0.8413 ± 0.0059	0.3536 ± 0.0940	0.3904 ± 0.2096
Switzerland	GCN (Ours)	116	0.5907 ± 0.0773	0.7286 ± 0.0628	0.7444 ± 0.0083	0.1384 ± 0.0482	0.6750 ± 0.0612
Switzerland	Logistic Regression	116	0.4556 ± 0.1819	0.5944 ± 0.1720	0.6287 ± 0.0195	0.0828 ± 0.0273	0.7000 ± 0.1696
Switzerland	Random Forest	116	0.8852 ± 0.1374	0.9042 ± 0.0751	0.6748 ± 0.0724	0.0315 ± 0.1200	0.2000 ± 0.3410
VA Long Beach	GCN (Ours)	140	0.7545 ± 0.0551	0.7994 ± 0.0229	0.7211 ± 0.0068	0.2451 ± 0.0700	0.5200 ± 0.1343
VA Long Beach	Logistic Regression	140	0.5309 ± 0.2310	0.6091 ± 0.1767	0.5697 ± 0.0055	0.0545 ± 0.0120	0.5267 ± 0.2332
VA Long Beach	Random Forest	140	0.9436 ± 0.1082	0.8606 ± 0.0420	0.6676 ± 0.0198	0.1160 ± 0.0661	0.1133 ± 0.1771

(b) Statistical comparison

Cohort	Comparison	AUC_{GCN}	AUC_{other}	ΔAUC	DeLong p	McNemar p	Holm p (DeLong)	Sig. Holm (DeLong)	Holm p (McNemar)	Sig. Holm (McNemar)
Hungarian	GCN vs Logistic Regression	0.8737	0.7046	0.1691	5.1e-08	1.4e-05	0.0	yes	0.0001	yes
Hungarian	GCN vs Random Forest	0.8737	0.8402	0.0335	0.1459	4.72e-10	0.3801	no	0.0	yes
Switzerland	GCN vs Logistic Regression	0.7407	0.5914	0.1493	0.3626	1.96e-05	0.3801	no	0.0001	yes
Switzerland	GCN vs Random Forest	0.7407	0.6152	0.1256	0.02	1.3e-09	0.1	no	0.0	yes
VA Long Beach	GCN vs Logistic Regression	0.7255	0.5636	0.1618	0.0219	0.2807	0.1	no	0.2807	no
VA Long Beach	GCN vs Random Forest	0.7255	0.6385	0.087	0.1267	0.0489	0.3801	no	0.0978	no

artefact of one explainer. The errors that remain concentrate near the decision boundary (Section VII-K), which is the desirable failure profile for a decision aid that defers uncertain cases rather than one that fails confidently.

P2: Deployability. Because each patient is an independent graph, inference is inductive by construction: the topology learned on the development cohort is applied unchanged, and a single patient can be scored with no cohort present. This is not a theoretical nicety. The conventional patient-similarity GNN we implement as a baseline must first construct a k-nearest-neighbour graph over whatever cohort it is asked to score (Section VII-C); it therefore cannot score an isolated referral at all, and its predictions for a given patient depend on which other patients happen to be present. The feature-node model carries none of that coupling: scoring patients individually reproduces batch predictions to 3×10^{-8} at 2.6 ms each, from an 11.4 KB model (Section VII-D). It is also the best-calibrated model out-of-fold and the one whose calibration best survives transport at realistic prevalence—a property that matters more than discrimination for a tool intended to report risk rather than rank patients. Where it does *not* yet earn its place is clinical utility on high-prevalence cohorts: on Switzerland and VA no

TABLE XX

Paired patient-level bootstrap on the external cohorts. Panel (a) reports per-model ROC-AUC with percentile intervals; panel (b) reports the paired differences against the feature-node model.

(a) Bootstrap ROC-AUC

Cohort	Model	n (pos/neg)	AUC	95% CI	CI width
Hungarian	GCN (Ours)	105/187	0.875	[0.829, 0.918]	0.088
Hungarian	Logistic Regression	105/187	0.706	[0.642, 0.767]	0.125
Hungarian	Random Forest	105/187	0.841	[0.786, 0.891]	0.105
Switzerland	GCN (Ours)	108/8	0.745	[0.543, 0.927]	0.384
Switzerland	Logistic Regression	108/8	0.628	[0.392, 0.833]	0.441
Switzerland	Random Forest	108/8	0.678	[0.430, 0.897]	0.467
VA Long Beach	GCN (Ours)	110/30	0.722	[0.622, 0.813]	0.191
VA Long Beach	Logistic Regression	110/30	0.571	[0.455, 0.688]	0.233
VA Long Beach	Random Forest	110/30	0.669	[0.553, 0.772]	0.219

(b) Paired differences

Cohort	Comparison	ΔAUC	95% CI	Excludes zero
Hungarian	GCN − Logistic Regression	0.169	[+0.108, +0.231]	yes
Hungarian	GCN − Random Forest	0.034	[-0.007, +0.075]	no
Switzerland	GCN − Logistic Regression	0.116	[-0.184, +0.407]	no
Switzerland	GCN − Random Forest	0.067	[-0.091, +0.215]	no
VA Long Beach	GCN − Logistic Regression	0.151	[+0.020, +0.289]	yes
VA Long Beach	GCN − Random Forest	0.052	[-0.032, +0.156]	no

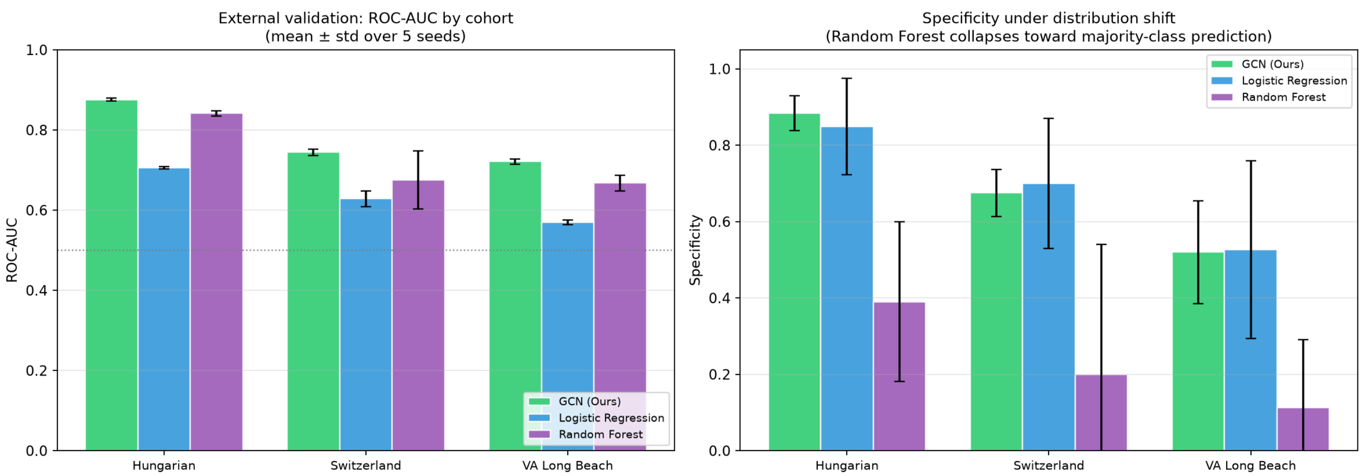

Fig. 12. Cross-cohort transport evaluation. Left: ROC-AUC by cohort for the transported models (mean±std over five seeds); the feature-node GCN leads on all three cohorts. Right: specificity under distribution shift; Random Forest degenerates toward majority-class prediction while the GCN retains a balanced operating point.

model improves on treating every patient, so our demonstrated utility rests on Hungarian alone.

P3: Transportability. Fitting every component on Cleveland and touching each external cohort once, the feature-node model attains the higher ROC-AUC in all six model–cohort comparisons, though paired bootstrap intervals establish this only against Logistic Regression and only on two cohorts (Section VII-L). The more robust signal is the difference in *failure mode*: Random Forest drifts toward majority-class prediction under shift, its specificity falling to 0.39, 0.20 and 0.11, and the patient-similarity GNN collapses entirely on VA, labelling every patient diseased, whereas the feature-node model retains a usable operating point throughout. We offer a mechanism: a model that must rebuild its graph from the target cohort inherits that cohort's distributional distortion into the very structure over which messages pass, whereas a fixed feature topology derived from the development cohort does not. That mechanism makes a testable prediction—degrading the topology should hurt transport while leaving in-distribution accuracy untouched—and Section VII-M provides suggestive evidence consistent with it: a fully connected graph transports worse despite being equally connected, implicating *selective* sparsity rather than connectivity alone. The effect is modest and, at the patient level, established only on the best-powered cohort, with no pairwise contrast surviving Holm correction of the DeLong tests; it is nonetheless the most direct evidence we have that the graph is doing work, and evidence no in-distribution experiment in this paper could have produced.

What the evidence does not support. Two claims we might have made, and do not. First, accuracy: the model is statistically indistinguishable from all six classical baselines in-distribution, including the two modern gradient-boosted-tree models (XGBoost, LightGBM), and we report that parity as a finding rather than minimising it. Second, architecture:

substituting GAT, GraphSAGE or GIN changes no metric significantly, so the contribution cannot be attributed to the convolution. What remains, and what the ablation localises, is the graph *construction*—selective, correlation-derived sparsity, which outperforms both random and fully connected topologies of comparable size. More broadly, our results caution that on small, near-balanced clinical cohorts differences between well-tuned models are usually within noise. This matters for interpreting the many prior studies on this same cohort that report high headline accuracies from hybrid or ensemble pipelines without paired significance testing across a shared protocol [42], [43]: without such testing it cannot be established whether the reported margins exceed split-to-split variation. Only a strictly shared protocol with significance testing distinguishes genuine improvement from favourable splitting, and reporting a model's non-superiority is often the more informative result. Table XXI positions our formulation against representative paradigms.

IX. Limitations

(i) The Cleveland cohort is small ($n = 303$) and single-centre, and the near-ceiling AUC leaves little separation between models, so accuracy parity with strong baselines is expected. *(ii)* Each node carries a single scalar feature, which provably limits the expressive advantage of message passing: the layer-1 node representation is rank-one in direction by construction and reaches an effective rank of only 1.73 out of 13 after two layers (Section VII-I). Consistent with this, the graph does not significantly outperform independent feature processing, so its principal value is representational and explanatory rather than predictive. The obvious remedy—richer node features—is not straightforward: a learnable per-feature identity embedding raises the effective rank only when the graph is removed, and with edges present message passing smooths the added identity away and degrades accuracy monotonically with edge density. Lifting this ceiling therefore requires an aggregation scheme that preserves node identity, not merely a wider node encoding, and we identify that as the open problem rather than a routine extension. *(iii)* Pearson correlation captures only linear feature dependence and may miss nonlinear interactions; mutual-information or copula-based graphs are promising alternatives. *(iv)* The MST guarantees connectivity but does not improve internal accuracy: at the operating threshold it adds no edges at all. Under distribution shift an MST-supported sparse topology is associated with better transport (Section VII-M), but that evidence is limited—the effect is modest, established at the patient level only on the best-powered cohort, and no pairwise contrast survives correction for multiple comparisons—so we treat transportability as directionally supported rather than established. The construction rule is likewise not empirically decisive on this cohort: a global correlation cut-off and a per-node k-NN rule of comparable density are indistinguishable after correction (Section VII-H), so our preference rests on parsimony and interpretability rather than measured performance. *(v)* The feature graph is small (thirteen nodes), bounding the depth of useful message passing. *(vi)* No temporal or longitudinal information is modelled. *(vii)* Cross-site validation is necessarily restricted to an eight-feature model, because `ca`, `thal` and `slope` are not recorded in the other cohorts; the full 13-feature model reported in Section VII therefore remains validated only internally. Moreover, the Hungarian, Switzerland and VA cohorts are not independent external data in the strongest sense: they are constituent databases of the same UCI Heart Disease collection [24], [25] as Cleveland, sharing variable definitions and recruitment era, so this evaluation demonstrates cross-site robustness rather than validation against an unrelated population (Section VII-L). *(viii)* Two of the three external cohorts are severely imbalanced (eight and thirty negative cases), so their estimates carry very wide uncertainty—the bootstrap 95% interval for the Switzerland AUC spans $[0.543, 0.927]$—and no comparison against Random Forest excludes zero; where DeLong's test and the bootstrap disagree we defer to the bootstrap, since asymptotic tests are unreliable with eight negative cases. *(ix)* The advantage over the patient-similarity formulation is partial rather than uniform: in-distribution it reaches significance in only one of three seeds, and on transport that model attains a marginally higher AUC on Switzerland, so our claim rests on the difference in failure mode and on the inductive/transductive distinction, not on a uniform margin. *(x)* We compared four message-passing operators but did not tune each one's hyper-parameters separately. *(xi)* Decision-curve analysis establishes net benefit over the treat-all strategy on Hungarian only; on Switzerland (93% prevalence) and VA (79%) no model, ours included, improves on treating every patient, so our clinical-utility evidence rests on a single external cohort. *(xii)* Calibration was assessed but not corrected; recalibration would likely improve the poorly calibrated external estimates.

X. Conclusion

When candidate models on a small clinical cohort are statistically tied on accuracy—as they are here—the decision between them should turn on representation, deployability and transportability, and we evaluated a feature-node graph formulation against exactly those criteria. The formulation converts a modelling assumption into an inspectable topology that is validated spectrally before training and explained by three concordant attribution methods; it is inductive by construction, scoring an isolated patient from an 11.4 KB, 1,281-parameter model where the patient-similarity baseline cannot score such a patient at all; and it attains the higher ROC-AUC in all six external comparisons while retaining a usable operating point where the baselines drift toward majority-class prediction. We bound both claims honestly: the transport advantage is established only against Logistic Regression and only on two cohorts, and on accuracy we report a null result and decline to dress it otherwise. Substituting GAT, GraphSAGE or GIN changes no metric significantly, which localises whatever contribution exists to the graph construction rather than the operator. We hope the honest positioning and full reproducibility of this study serve as a useful template for evaluating graph-based models on small clinical datasets, where the temptation to report a fractional accuracy gain

TABLE XXI
Qualitative comparison with representative paradigms.

Paradigm	Reported accuracy regime	Explainability	Inference mode	Interpretability notes
Classical tabular ML [42], [43]	High	None (post-hoc only)	N/A	Transparent at coefficient/importance level
Patient-similarity GNN [9]	Moderate–High	Node/edge masks on the population graph	Transductive	Explains via cohort neighbours, not features
Feature-node GCN (Ours)	Comparable to classical ML	GNNExplainer + Integrated Gradients + Saliency (8 metrics)	Inductive	Exposes feature-level interaction structure

is strong and the evidence for it is usually absent. Future work will pursue richer node features, nonlinear dependence graphs, calibrated risk estimates, and prospective multi-centre validation on cohorts that record the full feature set.

References

[1] World Health Organization, "Cardiovascular diseases (cvds) fact sheet," World Health Organization, Geneva, Switzerland, Tech. Rep., 2021.

[2] J. Petch, S. Di, and W. Nelson, "Opening the black box: The promise and limitations of explainable machine learning in cardiology," *Canadian Journal of Cardiology*, vol. 38, no. 2, pp. 204–213, 2022.

[3] M. A. Ahmad, C. Eckert, and A. Teredesai, "Interpretable machine learning in healthcare," *IEEE Intelligent Informatics Bulletin*, 2018.

[4] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," in *International Conference on Learning Representations (ICLR)*, 2017.

[5] J. Zhou, G. Cui, S. Hu, Z. Zhang, C. Yang, Z. Liu, L. Wang, C. Li, and M. Sun, "Graph neural networks: A review of methods and applications," *AI Open*, vol. 1, pp. 57–81, 2020.

[6] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Liò, and Y. Bengio, "Graph attention networks," in *International Conference on Learning Representations (ICLR)*, 2018.

[7] W. L. Hamilton, R. Ying, and J. Leskovec, "Inductive representation learning on large graphs," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

[8] K. Xu, W. Hu, J. Leskovec, and S. Jegelka, "How powerful are graph neural networks?" in *International Conference on Learning Representations (ICLR)*, 2019.

[9] S. Parisot, S. I. Ktena, E. Ferrante, M. Lee, R. Guerrero, B. Glocker, and D. Rueckert, "Disease prediction using graph convolutional networks: Application to autism spectrum disorder and alzheimer's disease," *Medical Image Analysis*, vol. 48, pp. 117–130, 2018.

[10] M. M. Li, K. Huang, and M. Zitnik, "Graph representation learning in biomedicine and healthcare," *Nature Biomedical Engineering*, vol. 6, no. 12, pp. 1353–1369, 2022.

[11] Y. Ektefaie, G. Dasoulas, A. Noori, M. Farhat, and M. Zitnik, "Multimodal learning with graphs," *Nature Machine Intelligence*, vol. 5, no. 4, pp. 340–350, 2023.

[12] J. Yan, J. Chen, Y. Wu, D. Z. Chen, and J. Wu, "T2G-FORMER: Organizing tabular features into relation graphs promotes heterogeneous feature interaction," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 37, no. 9, 2023, pp. 10 720–10 728.

[13] M. Villaizán-Vallelado, M. Salvatori, B. Carro, and A. J. Sanchez-Esguevillas, "Graph neural network contextual embedding for deep learning on tabular data," *Neural Networks*, vol. 173, p. 106180, 2024.

[14] C.-T. Li, Y.-C. Tsai, C.-Y. Chen, and J. C. Liao, "Graph neural networks for tabular data learning: A survey with taxonomy and directions," *ACM Computing Surveys*, vol. 58, no. 1, pp. 1–51, 2025.

[15] G. Badaro, M. Saeed, and P. Papotti, "Transformers for tabular data representation: A survey of models and applications," *Transactions of the Association for Computational Linguistics*, vol. 11, pp. 227–249, 2023.

[16] N. Hollmann, S. Müller, L. Purucker, A. Krishnakumar, M. Körfer, S. B. Hoo, R. T. Schirrmeister, and F. Hutter, "Accurate predictions on small data with a tabular foundation model," *Nature*, vol. 637, no. 8045, pp. 319–326, 2025.

[17] W. Samek, G. Montavon, S. Lapuschkin, C. J. Anders, and K.-R. Müller, "Explaining deep neural networks and beyond: A review of methods and applications," *Proceedings of the IEEE*, vol. 109, no. 3, pp. 247–278, 2021.

[18] P. Lisbôa, S. Saralajew, A. Vellido, R. Fernández-Domenech, and T. Villmann, "The coming of age of interpretable and explainable machine learning models," *Neurocomputing*, vol. 535, pp. 25–39, 2023.

[19] A. V. Ponce Bobadilla, V. Schmitt, C. S. Maier, S. Mensing, and S. Stodtmann, "Practical guide to SHAP analysis: Explaining supervised machine learning model predictions in drug development," *Clinical and Translational Science*, vol. 17, no. 11, p. e70056, 2024.

[20] R. Ying, D. Bourgeois, J. You, M. Zitnik, and J. Leskovec, "GNNExplainer: Generating explanations for graph neural networks," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

[21] M. Sundararajan, A. Taly, and Q. Yan, "Axiomatic attribution for deep networks," in *International Conference on Machine Learning (ICML)*, 2017.

[22] K. Simonyan, A. Vedaldi, and A. Zisserman, "Deep inside convolutional networks: Visualising image classification models and saliency maps," in *ICLR Workshop*, 2014.

[23] C.-K. Yeh, C.-Y. Hsieh, A. S. Suggala, D. I. Inouye, and P. Ravikumar, "On the (in)fidelity and sensitivity of explanations," *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

[24] A. Janosi, W. Steinbrunn, M. Pfisterer, and R. Detrano, "Heart Disease," UCI Machine Learning Repository, 1988, dOI: 10.24432/C52P4X.

[25] R. Detrano, A. Jánosi, W. Steinbrunn, M. Pfisterer, J.-J. Schmid, S. Sandhu, K. H. Guppy, S. Lee, and V. Froelicher, "International application of a new probability algorithm for the diagnosis of coronary artery disease," *The American Journal of Cardiology*, vol. 64, no. 5, pp. 304–310, 1989.

[26] R. C. Prim, "Shortest connection networks and some generalizations," *The Bell System Technical Journal*, vol. 36, no. 6, pp. 1389–1401, 1957.

[27] M. Fiedler, "Algebraic connectivity of graphs," *Czechoslovak Mathematical Journal*, vol. 23, no. 2, pp. 298–305, 1973.

[28] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, 2016, pp. 785–794.

[29] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, "LightGBM: A highly efficient gradient boosting decision tree," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017, pp. 3146–3154.

[30] B. W. Matthews, "Comparison of the predicted and observed secondary structure of T4 phage lysozyme," *Biochimica et Biophysica Acta (BBA) - Protein Structure*, vol. 405, no. 2, pp. 442–451, 1975.

[31] Q. McNemar, "Note on the sampling error of the difference between correlated proportions or percentages," *Psychometrika*, vol. 12, no. 2, pp. 153–157, 1947.

[32] F. Wilcoxon, "Individual comparisons by ranking methods," *Biometrics Bulletin*, vol. 1, no. 6, pp. 80–83, 1945.

[33] S. Holm, "A simple sequentially rejective multiple test procedure," *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979.

[34] A. Paszke *et al.*, "PyTorch: An imperative style, high-performance deep learning library," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

[35] M. Fey and J. E. Lenssen, "Fast graph representation learning with PyTorch Geometric," in *ICLR Workshop on Representation Learning on Graphs and Manifolds*, 2019.

[36] F. Pedregosa *et al.*, "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

[37] E. R. DeLong, D. M. DeLong, and D. L. Clarke-Pearson, "Comparing the areas under two or more correlated receiver operating characteristic curves: A nonparametric approach," *Biometrics*, vol. 44, no. 3, pp. 837–845, 1988.
[38] G. W. Brier, "Verification of forecasts expressed in terms of probability," *Monthly Weather Review*, vol. 78, no. 1, pp. 1–3, 1950.
[39] M. P. Naeini, G. F. Cooper, and M. Hauskrecht, "Obtaining well calibrated probabilities using Bayesian binning," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 29, no. 1, 2015, pp. 2901–2907.
[40] A. J. Vickers and E. B. Elkin, "Decision curve analysis: A novel method for evaluating prediction models," *Medical Decision Making*, vol. 26, no. 6, pp. 565–574, 2006.
[41] E. B. Page, "Ordered hypotheses for multiple treatments: A significance test for linear ranks," *Journal of the American Statistical Association*, vol. 58, no. 301, pp. 216–230, 1963.
[42] S. Mohan, C. Thirumalai, and G. Srivastava, "Effective heart disease prediction using hybrid machine learning techniques," *IEEE Access*, vol. 7, pp. 81 542–81 554, 2019.
[43] C. B. C. Latha and S. C. Jeeva, "Improving the accuracy of prediction of heart disease risk based on ensemble classification techniques," *Informatics in Medicine Unlocked*, vol. 16, p. 100203, 2019.

FIRST A. AUTHOR received the B.E. degree in computer science and engineering from Institution Name, City, Country, in 20xx, where they are currently pursuing the M.Tech. degree in artificial intelligence.

Their research interests include graph representation learning, explainable artificial intelligence, and machine learning for clinical decision support.